

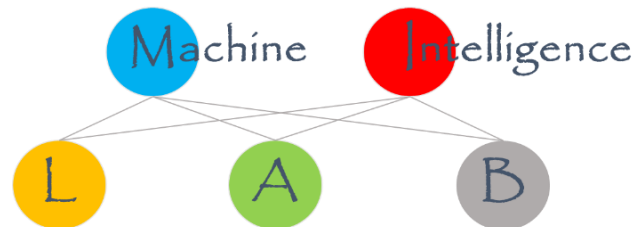


北京大學
PEKING UNIVERSITY

Visual Recognition – I

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Wangxuan Institute of Computer Technology
Peking University



**Several slides are adapted from related courses or tutorials.
Internal use only. Please do not distribute the slides.**

Outline

- **Visual Recognition: Task Definition and Challenges**
- **Bag-of-words Models**
 - interest point, local feature, quantization
- **(Locality): Spatial Pyramid Matching & Pyramid Match Kernel**
- **(Aggregation): VLAD, Fisher Vector, Soft Assignment**
- **(Dictionary): Vocabulary Tree, Sparse Coding**

Visual Recognition / Image Classification

- Give a binary label to indicate whether an object is present

Does this image contain a car?



cars



not cars



Semantic Gap

■ Images in the eyes of computer v.s. human

青海昆仑山南山口首次拍摄到雪豹影像

2021-03-25 11:03 中国新闻网

| 评论 | 分享 | 109



1 / 3

日前，青海省格尔木市摄影家协会沙漠拍客团队首次在距离市区40公里的昆仑山南山口，利用红外相机拍摄到了国家一级保护动物雪豹的高清活动影像，这也是格尔木地区首次发现并拍摄到清晰的雪豹影像。图为红外相机拍摄到的雪豹。

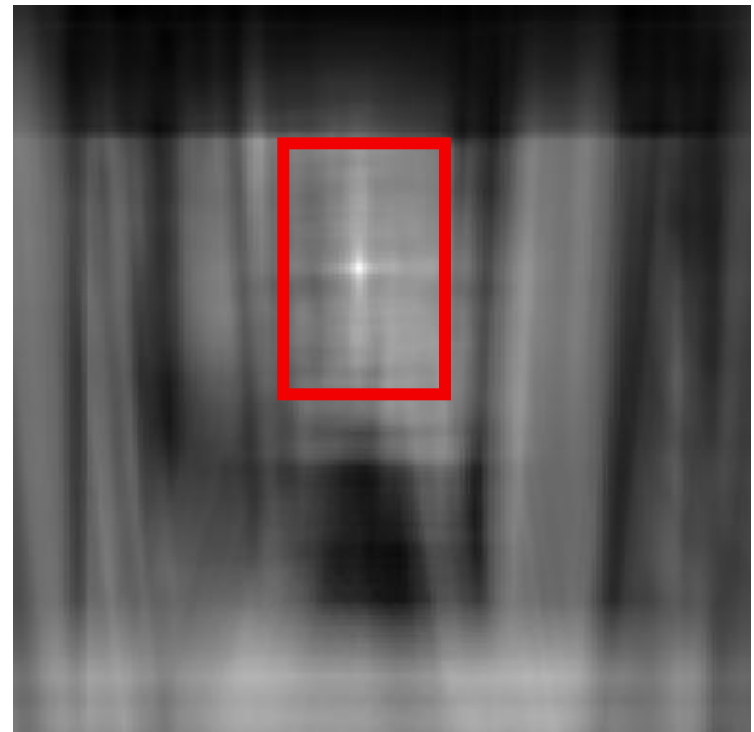
Semantic Gap

- Low effectiveness of visual features in representing semantics

Find the chair in this image



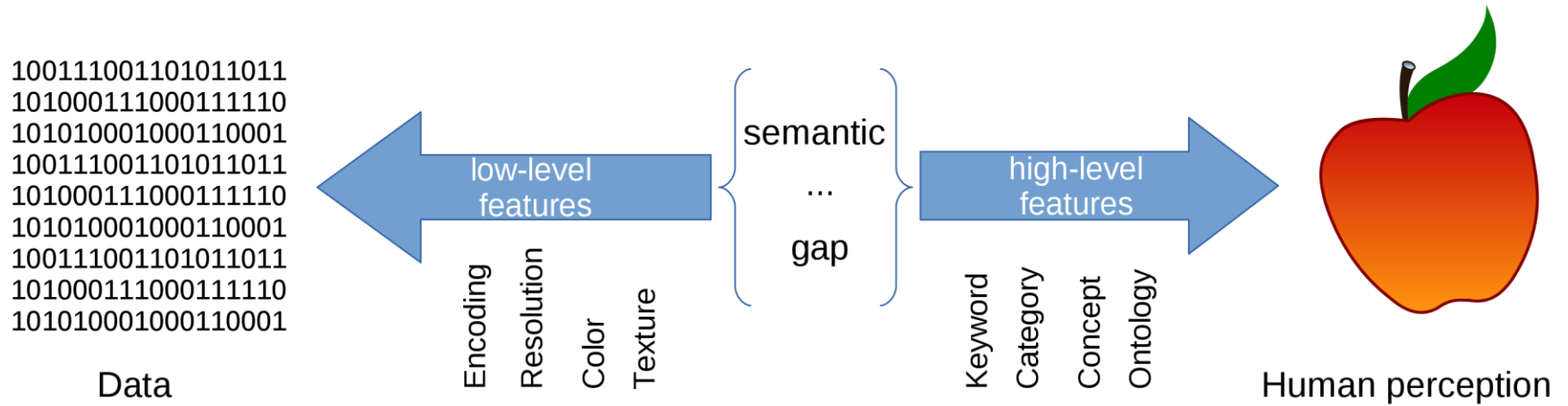
Output of normalized correlation



This is a chair

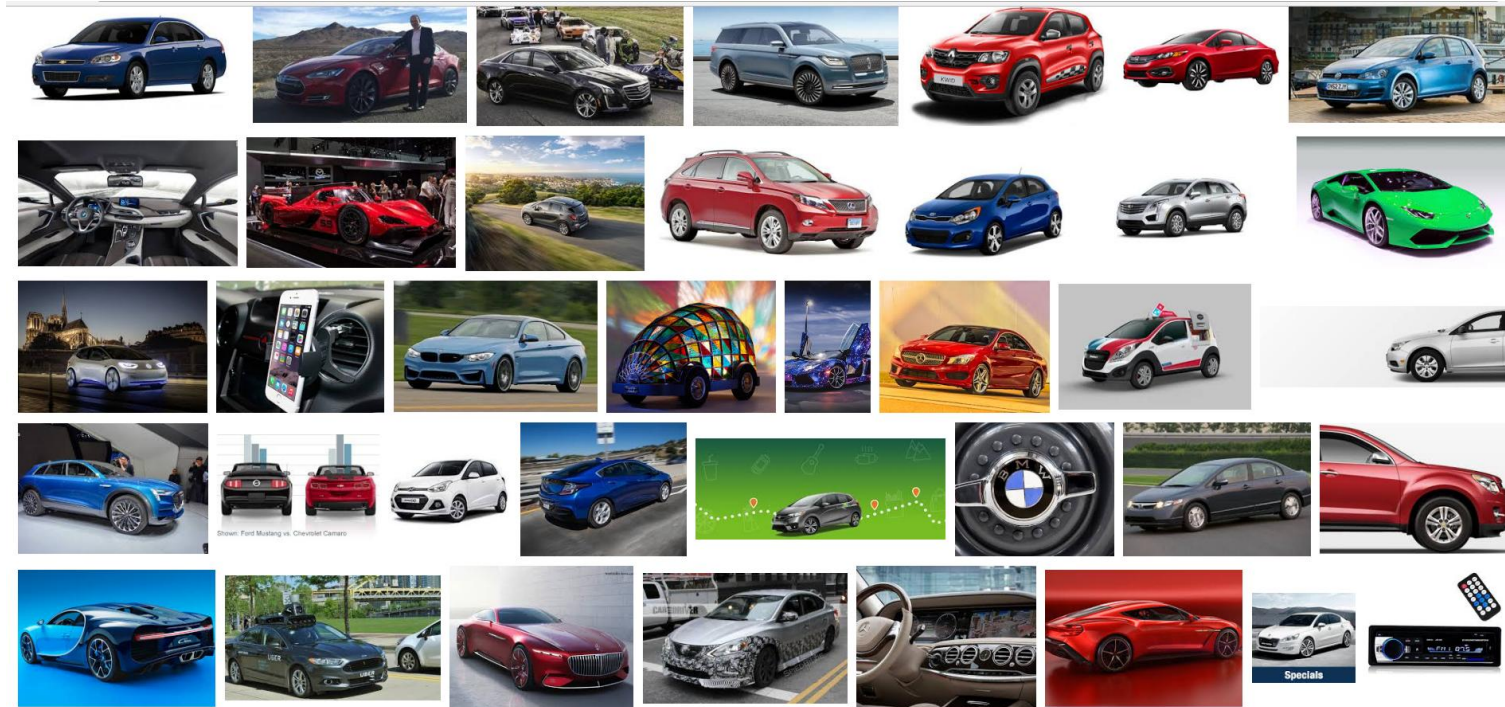


Semantic Gap



Visual Variations

- Pose, lighting, scale etc.



Granularity of Recognition



Category 1: Black-billed Cuckoo



Category 2: Mangrove Cuckoo



Category 3: Yellow-billed Cuckoo

Semantics in Context

- Recognizing complex concepts in context

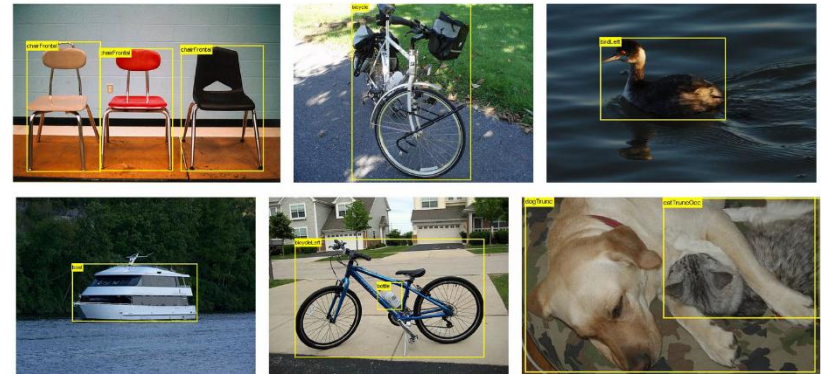


Image Classification @ CV Community

■ Pascal VOC Challenges 2005-2012

Table 1 The VOC classes

Vehicles	Household	Animals	Other
Aeroplane	Bottle	Bird	Person
Bicycle	Chair	Cat	
Boat	Dining table	Cow	
Bus	Potted plant	Dog	
Car	Sofa	Horse	
Motorbike	TV/Monitor	Sheep	
Train			



■ Large Scale Visual Recognition Challenge (ILSVRC) 2010-2016

1000 synsets for Object classification/localization

[kit fox, *Vulpes macrotis*](#)

[English setter](#)

[Australian terrier](#)

[grey whale, gray whale, devilfish, *Eschrichtius gibbosus*, *Eschrichtius robustus*](#)

[lesser panda, red panda, panda, bear cat, cat bear, *Ailurus fulgens*](#)

[Egyptian cat](#)

[ibex, *Capra ibex*](#)

[Persian cat](#)

[cougar, puma, catamount, mountain lion, painter, panther, *Felis concolor*](#)

[gazelle](#)

[porcupine, hedgehog](#)

[sea lion](#)

IMAGENET

14,197,122 images, 21841 synsets indexed

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ImageNet is an image database organized according to the WordNet hierarchy (currently only the nouns), in which each node of the hierarchy is depicted by hundreds and thousands of images. Currently we have an average of over five hundred images per node. We hope ImageNet will become a useful resource for researchers, educators, students and all of you who share our passion for pictures.

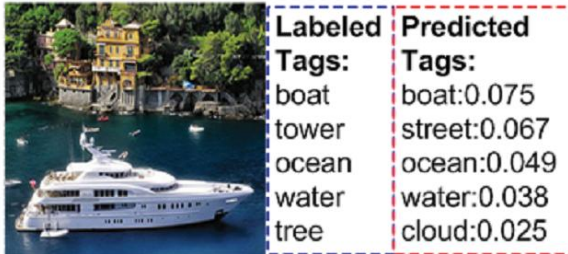
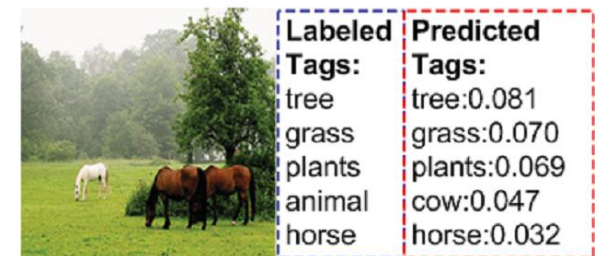
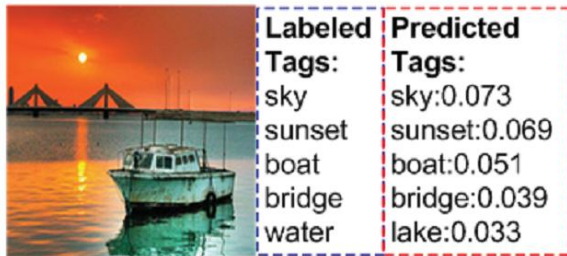
[Click here](#) to learn more about ImageNet, [Click here](#) to join the ImageNet mailing list.

<http://www.image-net.org/>

<http://www.image-net.org/challenges/LSVRC/2016/index>

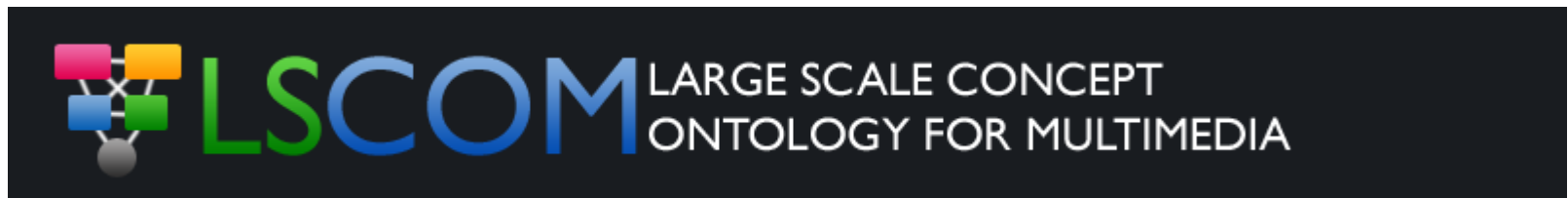
Image Classification @ Multimedia Community

- Image tagging: YFCC100M



- Concept detection: TRECVID SIN task

Ontology design -> Image collecting -> Image annotation -> Concept detection



Performance Criteria

- Scope: what kinds of objects can we accurately categorize?
- Generalization ability to unseen samples
- Efficiency
- Accuracy

Bag-of-Words Model

- A text (such as a sentence or a document) is represented as the bag (multiset) of its words, disregarding grammar and even word order but keeping multiplicity.
- a special case of n-gram in NLP

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!



it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1
...	...

“Visual” Bag-of-Words Model

Object

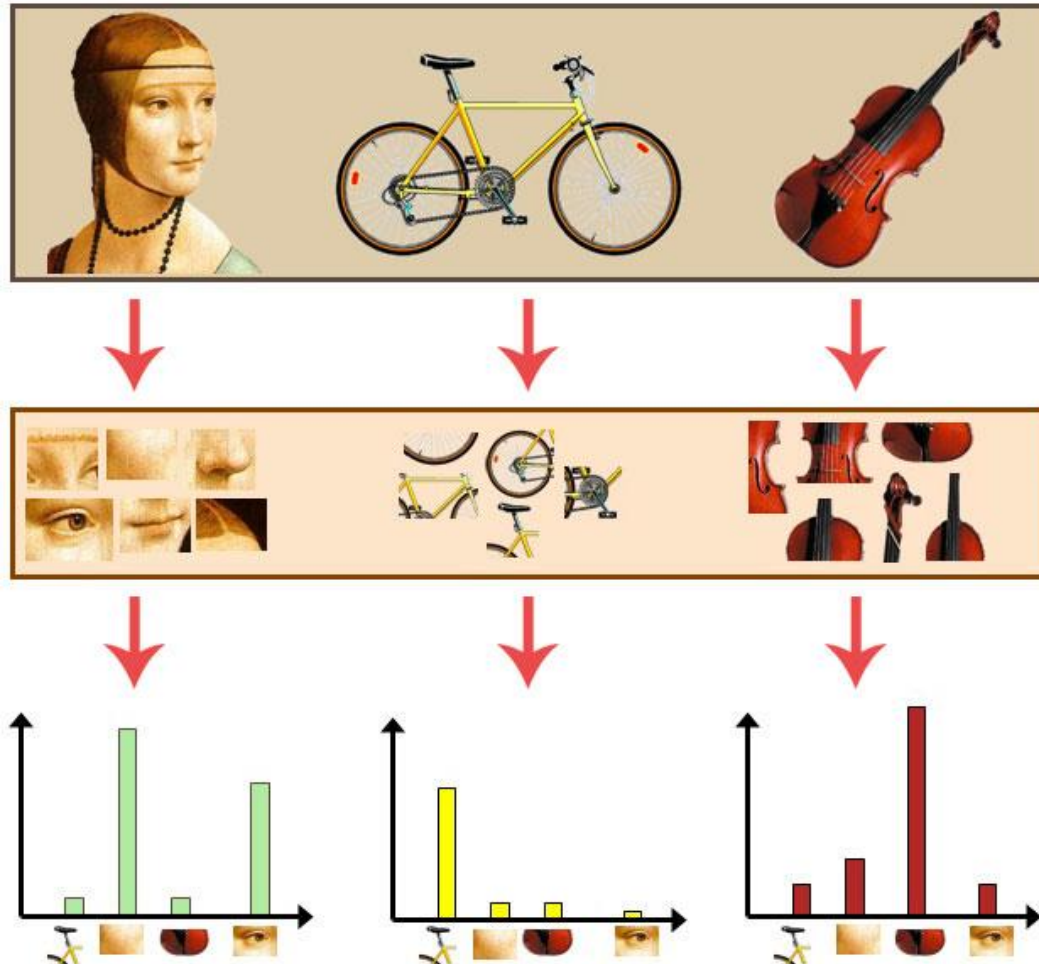


Bag of 'words'



“Visual” Bag-of-Words Model

- Use bags of “visual” words for image recognition or matching



“Visual” Bag-of-Words Model

- The two images below look similar. But, how to quantize the similarity?



Idea: similarity is local.

“Visual” Bag-of-Words Model

- Key elements in building visual bag-of-words model
 - Find “interest” points
 - Calculate local descriptors
 - Learn a visual dictionary and quantize each local descriptors
 - Calculate a histogram of all visual words in an image. Done!!



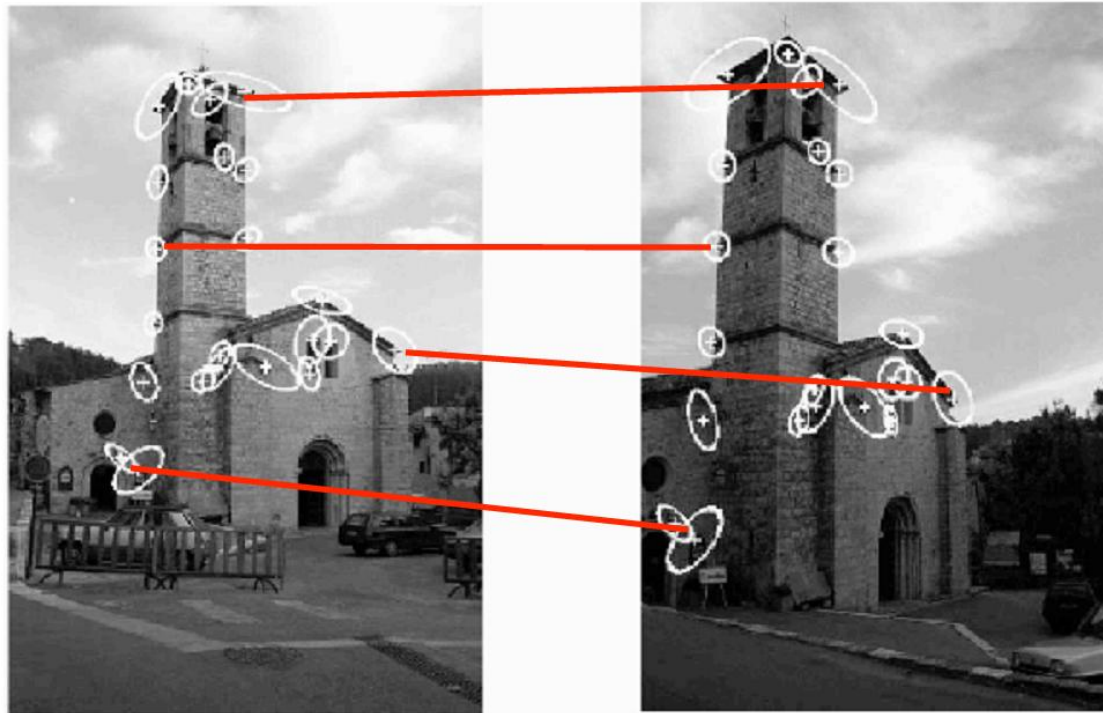
The First Step: Find the “Interest” Point

- Visual unique-ness



Harris Corner Detector

- The image case: Harris corner detector

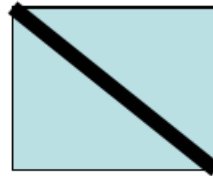


Harris Corner Detector

- What are good local point?



flat



edge

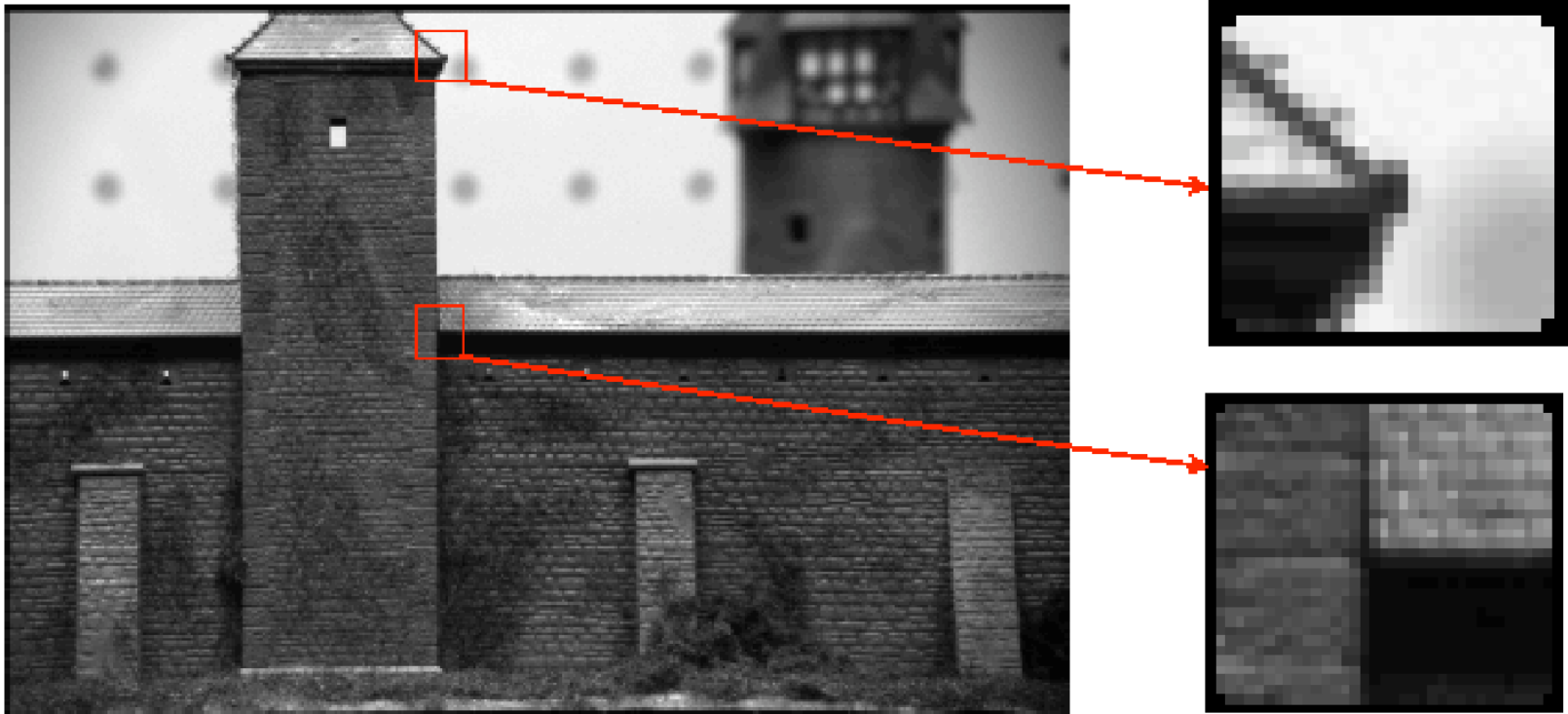


corner

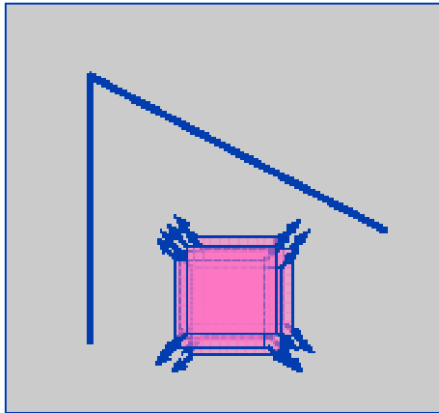


Harris Corner Detector

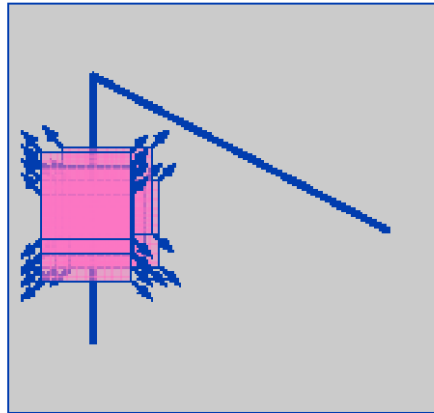
- What are good local point?
- Junction of contours / robust over illumination/viewpoint change / locally “unique”



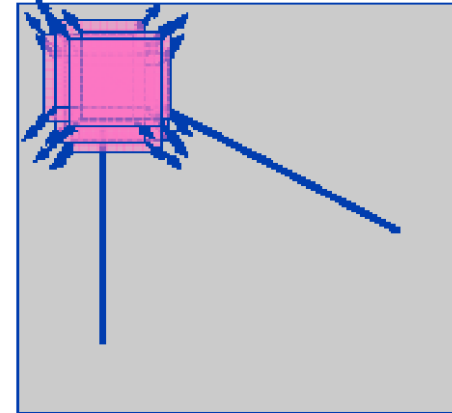
Harris Corner Detector



“flat” region:
no change in
all directions



“edge”:
no change along
the edge direction



“corner”:
significant change
in all directions

**Shifting the window in any direction should yield
a large change in appearance.**

Harris Corner Detector

Change of intensity for the shift $[u, v]$:

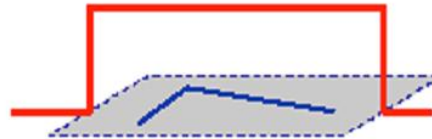
$$E(u, v) = \sum_{x, y} w(x, y) [I(x+u, y+v) - I(x, y)]^2$$

Window
function

Shifted
intensity

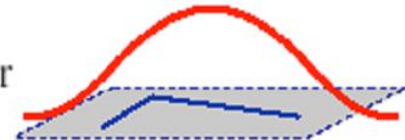
Intensity

Window function $w(x, y) =$



1 in window, 0 outside

or



Gaussian

Harris Corner Detector

Based on first order local approximation

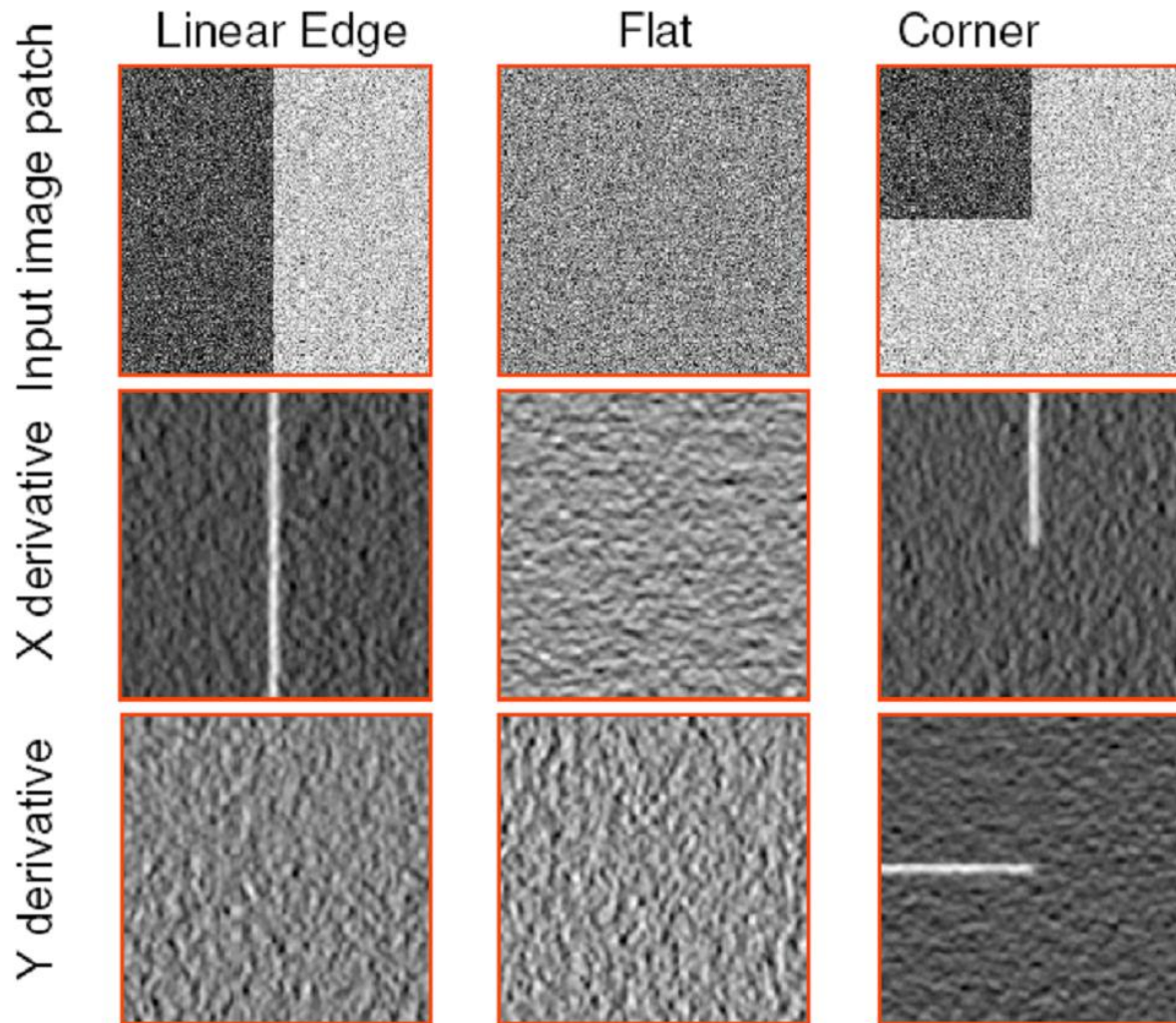
$$E(u, v) \approx \sum_{x,y} [I(x, y) + uI_x + vI_y - I(x, y)]^2$$

$$E(u, v) \approx \sum_{x,y} u^2 I_x^2 + 2uv I_x I_y + v^2 I_y^2$$

$$E(u, v) \approx [u \quad v] \left(\sum \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} \right) \begin{bmatrix} u \\ v \end{bmatrix}$$

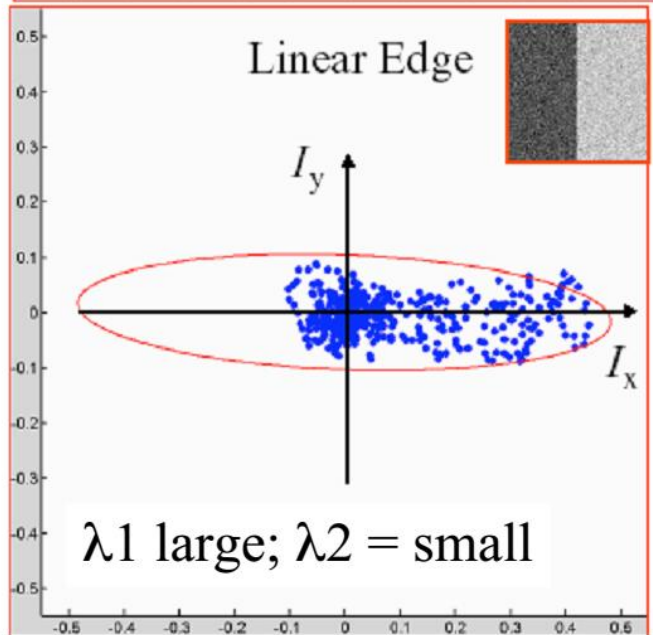
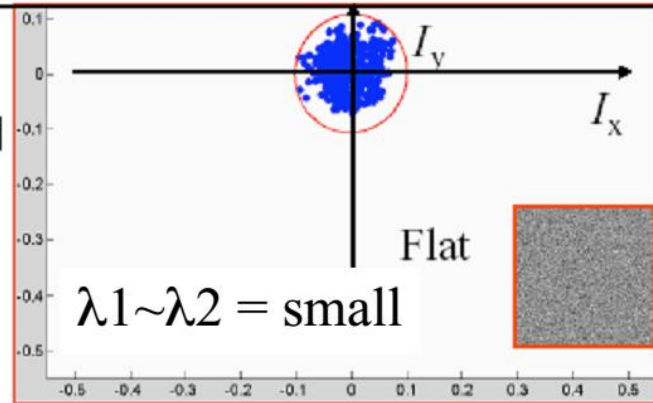
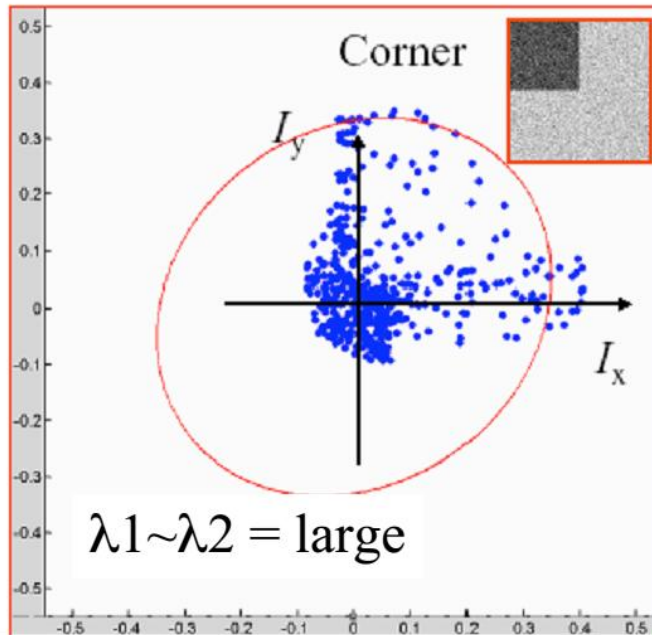
$$M = \sum w(x, y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} \quad E(u, v) \approx [u \quad v] M \begin{bmatrix} u \\ v \end{bmatrix}$$

Harris Corner Detector



Harris Corner Detector

The distribution of x and y derivatives can be characterized by the shape and size of the principal component ellipse

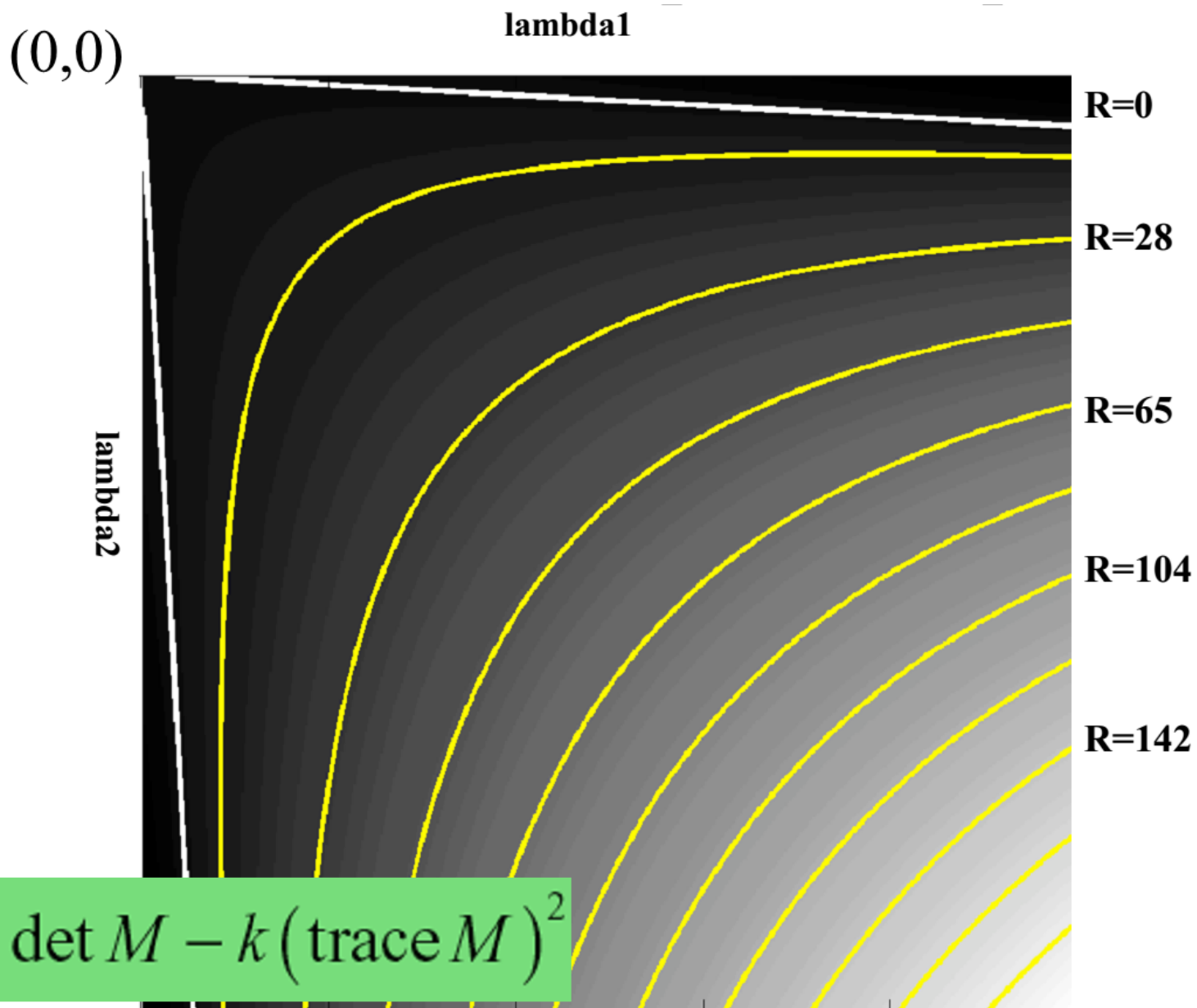


Harris Corner Detector

$$R = \det(M) - k(\text{trace}(M))^2$$

- $\det(M) = \lambda_1 \lambda_2$
- $\text{trace}(M) = \lambda_1 + \lambda_2$
- λ_1 and λ_2 are the eigen values of M

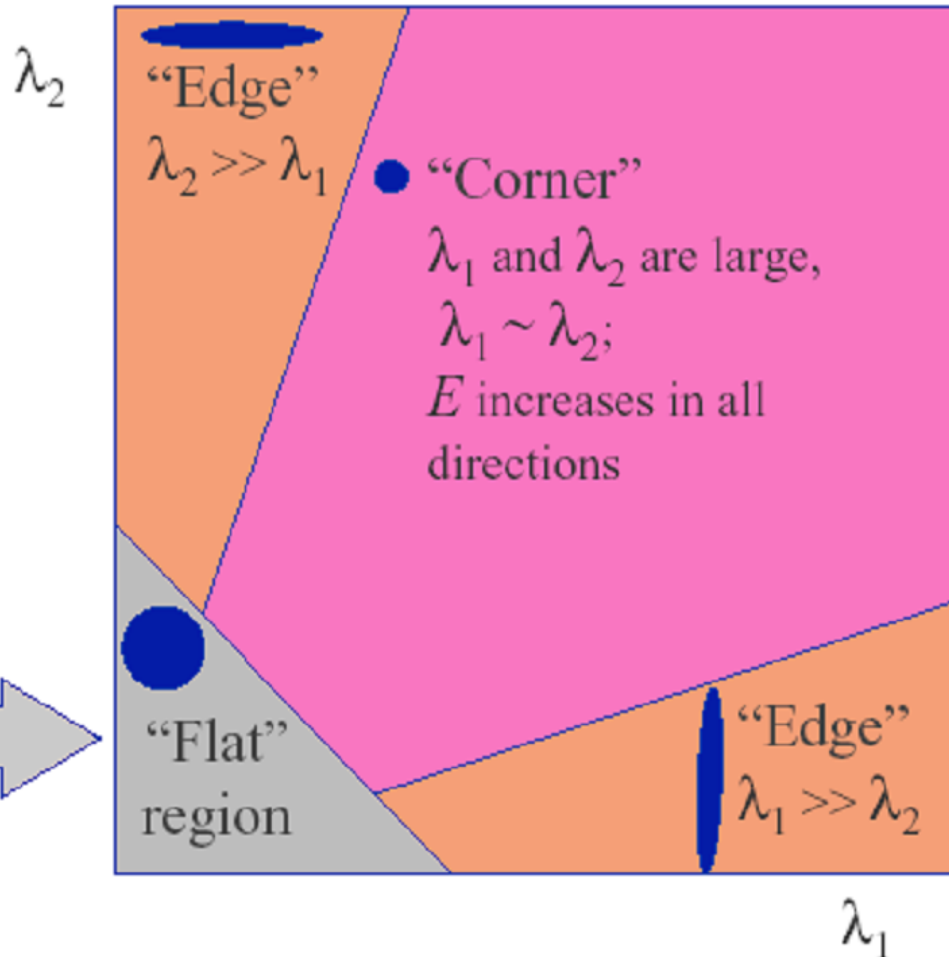
- When $|R|$ is small, which happens when λ_1 and λ_2 are small, the region is flat.
- When $R < 0$, which happens when $\lambda_1 \gg \lambda_2$ or vice versa, the region is edge.
- When R is large, which happens when λ_1 and λ_2 are large and $\lambda_1 \sim \lambda_2$, the region is a corner.



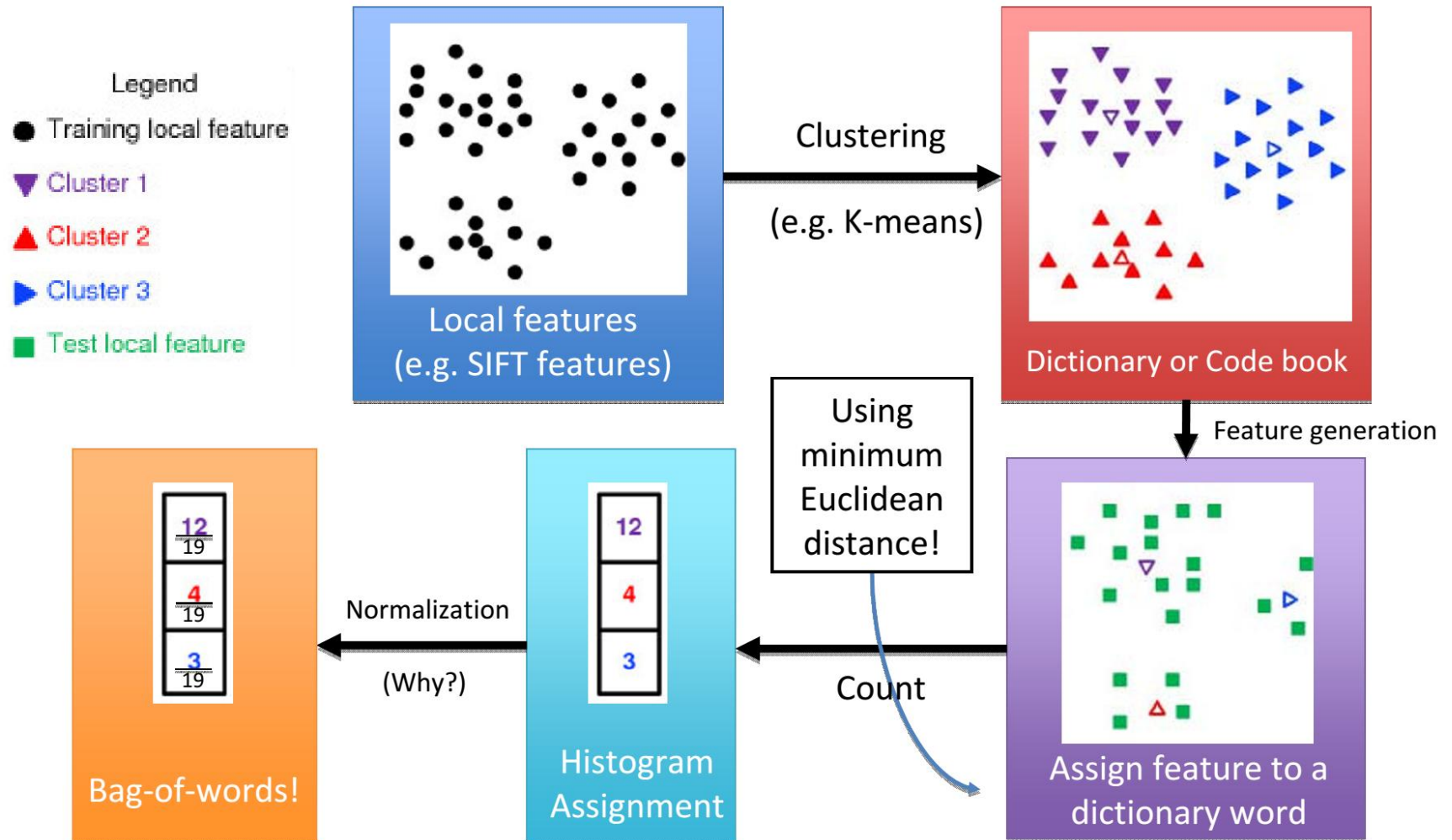
Harris Corner Detector

Classification of
image points using
eigenvalues of M :

λ_1 and λ_2 are small;
 E is almost constant
in all directions



Pipeline of Visual Bag-of-Words Model



Local Features – the “Words”

- SIFT (Scale-invariant feature transform)



<http://www.vlfeat.org/overview/sift.html>

- And its variants!

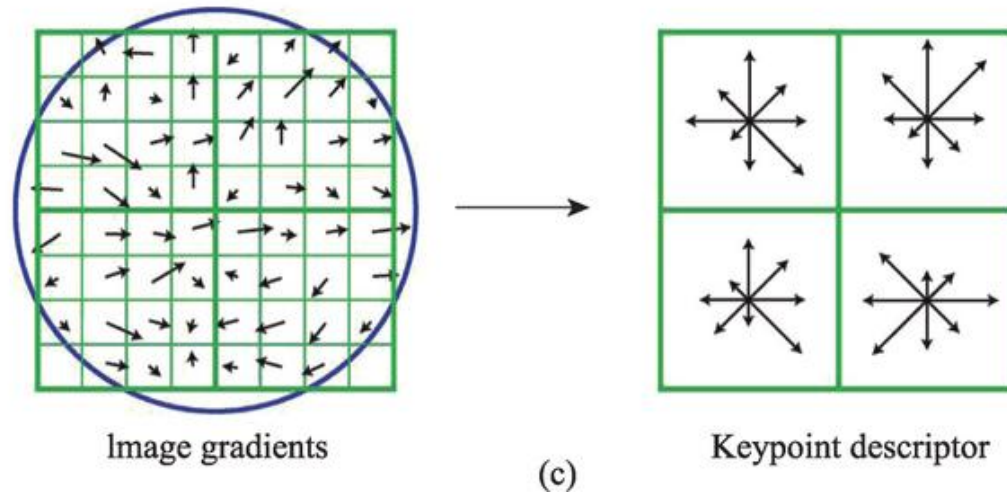
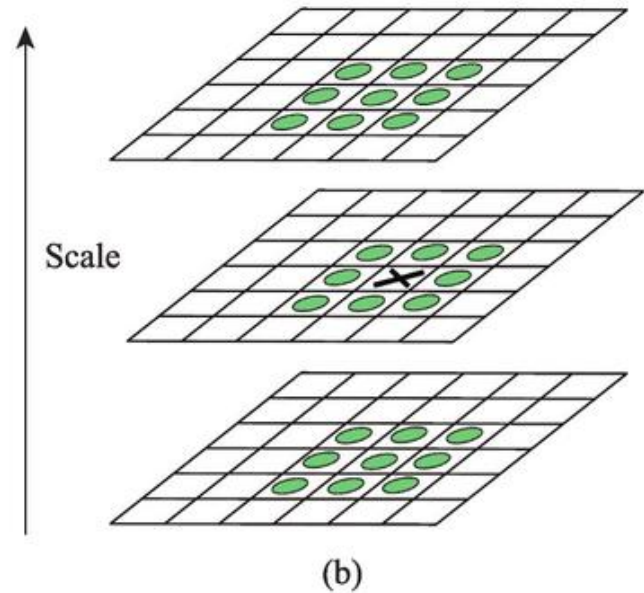
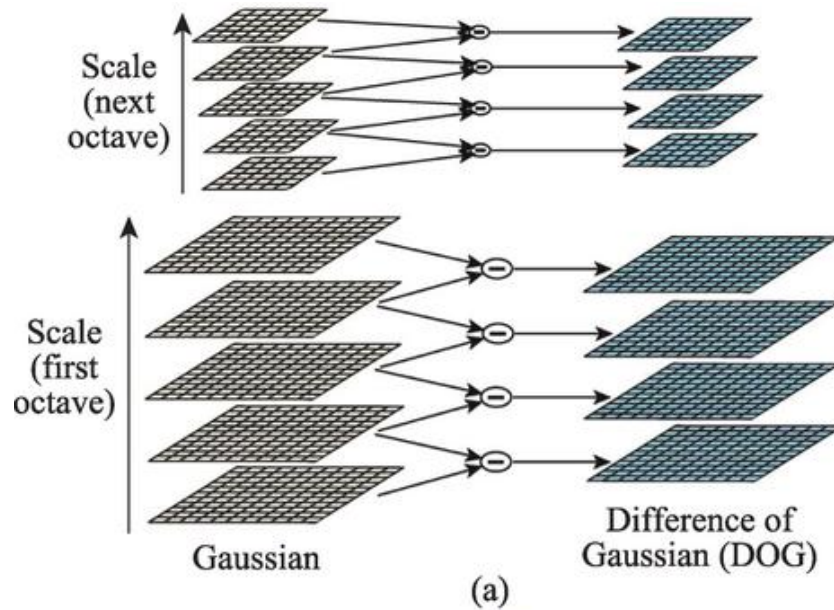
- SIFT (sift)
- HueSIFT (huesift)
- HSV-SIFT (hsvsift)
- OpponentSIFT (opponentsift)
- *rg*SIFT (rgsift)
- C-SIFT (csift)
- RGB-SIFT(rgbsift)

<http://koen.me/research/colordescriptors>

Dense SIFT

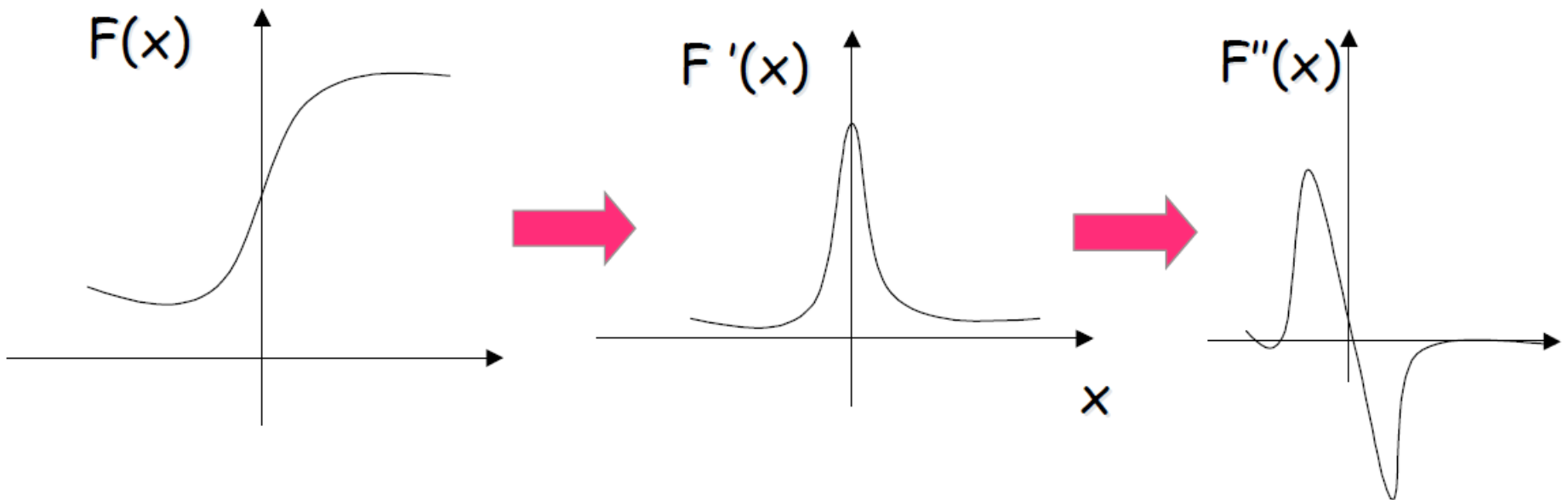


SIFT



First Derivative and Second Derivative

- peaks or valleys of the first derivative: sharp change of the image lightness, or zero-crossings of the second derivative



(1D example)

Numerical Calculation of Second Derivative and 2D Laplacian

■ Taylor Series expansion

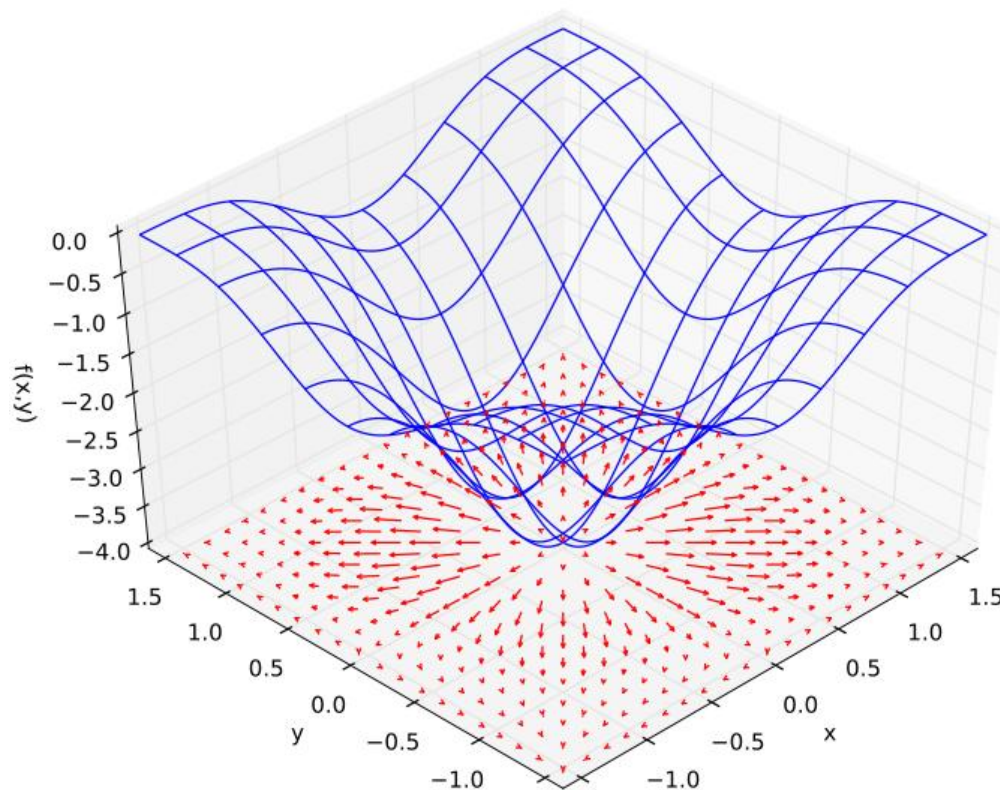
$$\begin{aligned}
 f(x+h) &= f(x) + hf'(x) + \frac{1}{2}h^2 f''(x) + \frac{1}{3!}h^3 f'''(x) + O(h^4) \\
 \text{+} \left[f(x-h) &= f(x) - hf'(x) + \frac{1}{2}h^2 f''(x) - \frac{1}{3!}h^3 f'''(x) + O(h^4) \right] \\
 \hline
 f(x+h) + f(x-h) &= 2f(x) + h^2 f''(x) + O(h^4) \\
 \frac{f(x-h) - 2f(x) + f(x+h)}{h^2} &= f''(x) + O(h^2)
 \end{aligned}
 \quad \Rightarrow \quad
 \begin{array}{|c|c|c|}
 \hline
 1 & -2 & 1 \\
 \hline
 \end{array}$$

■ 2D Laplacian

$$I_{xx} + I_{yy} = \left(\begin{bmatrix} 1 & -2 & 1 \end{bmatrix} + \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \right) * I = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}}_{\text{Laplacian filter } \nabla^2 \mathbf{I}(x,y)} * I$$

Geometric Interpretation of Laplacian

- Laplacian as blob detector (local maximum/minimum)



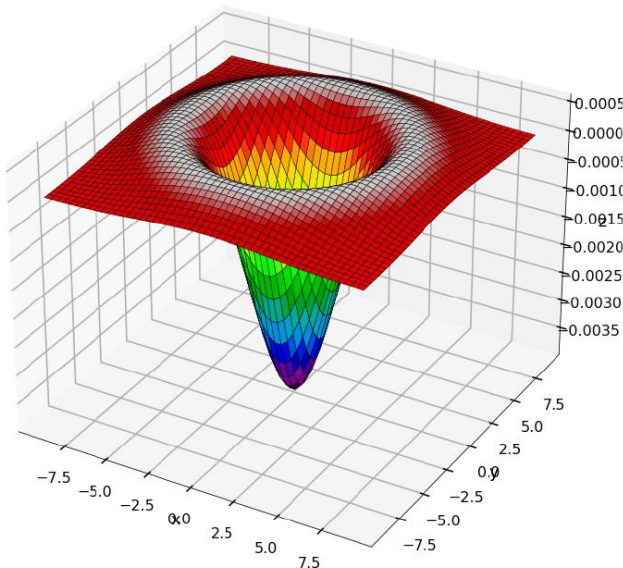
Approximating LoG with DoG

- Proof sketch

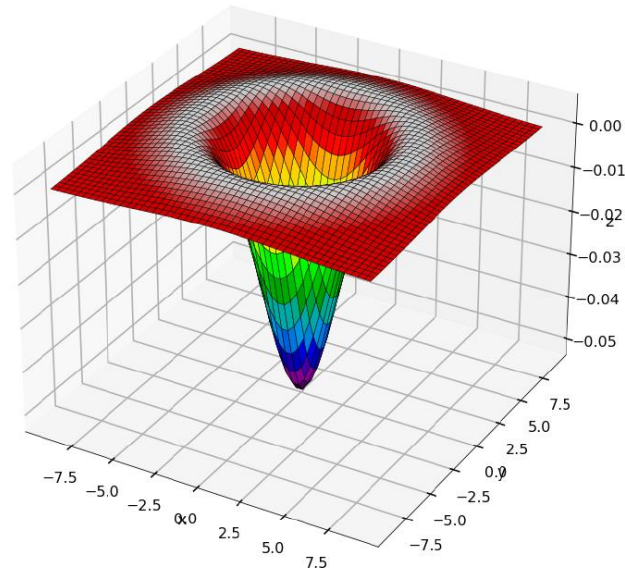
$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \quad \nabla^2 G(x, y) = G_{xx} + G_{yy} = \frac{1}{2\pi\sigma^4} e^{-\frac{x^2+y^2}{2\sigma^2}} \left(\frac{x^2 + y^2}{\sigma^2} - 2 \right)$$

$$LoG(x, y, \sigma^2) \propto \frac{\partial}{\partial (\sigma^2)} G(x, y, \sigma^2) = \lim_{h \rightarrow 0} \frac{G(x, y, \sigma^2 + h) - G(x, y, \sigma^2)}{h}$$

LoG ($\sigma=3$)

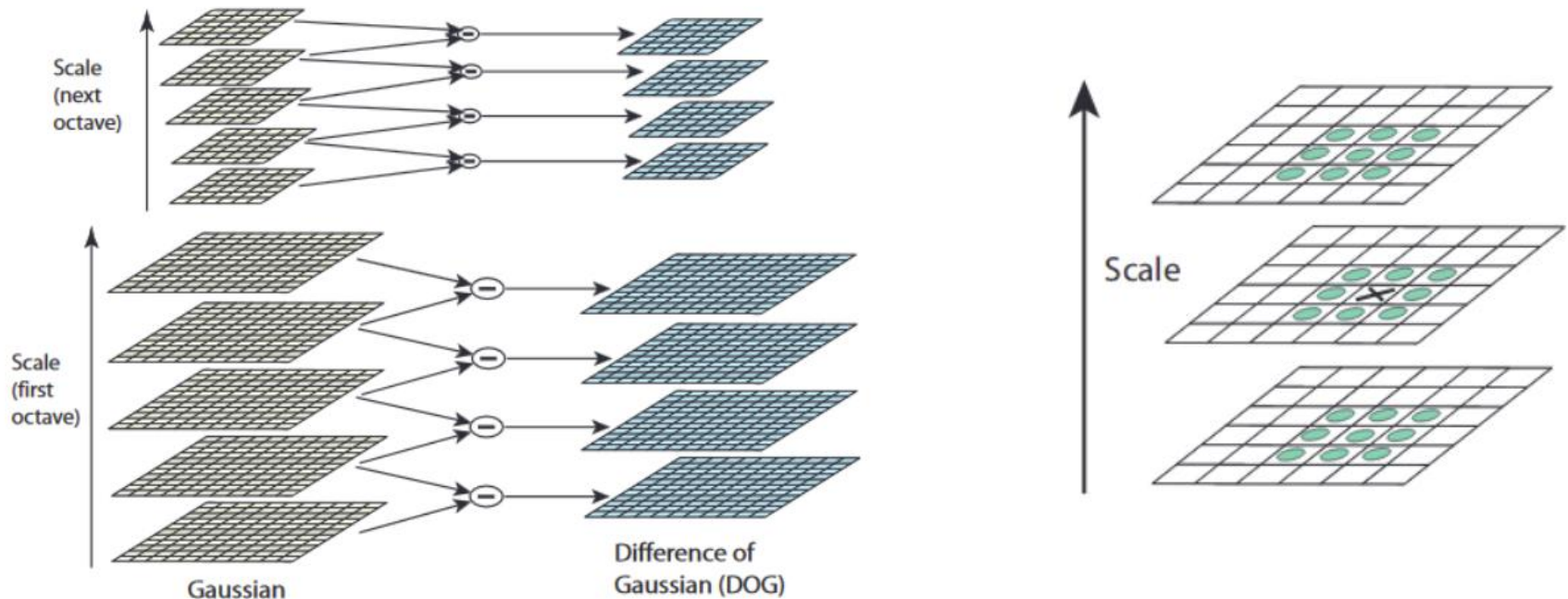


DoG ($\sigma_1 = 4.24, \sigma_2 = 2.12$)

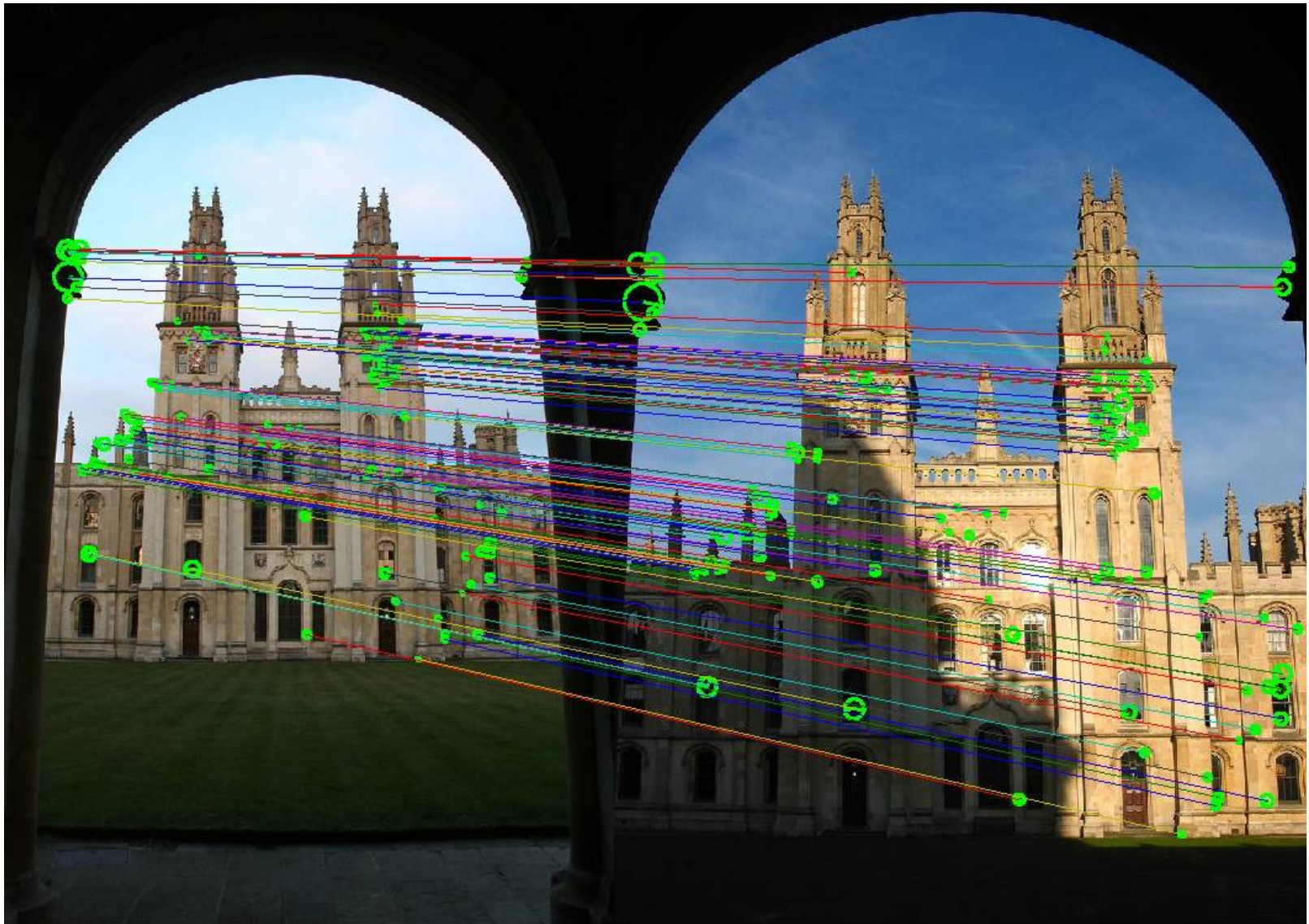


Finding interest points via DoG

- Locally maximum/minimum
- Remove edge points via Harris corner detector



SIFT for Cross-Scale Matching



Evaluation of Interest Point Detectors and Feature Descriptors for Visual Tracking

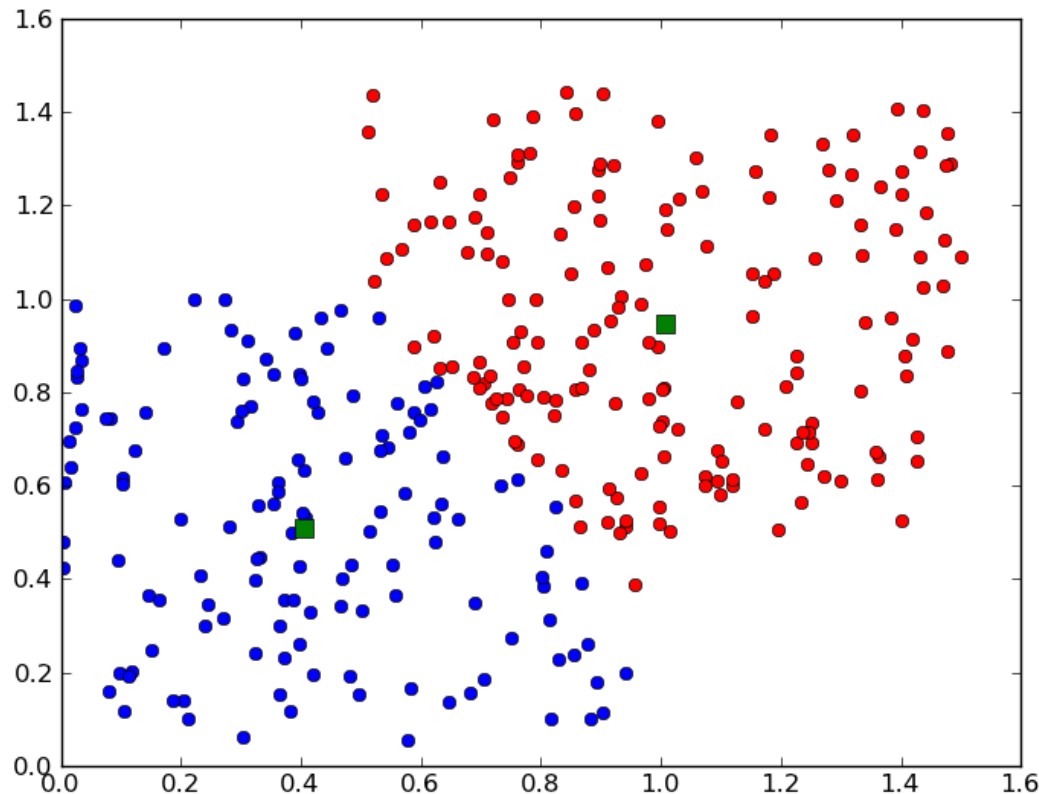
Steffen Gauglitz · Tobias Höllerer · Matthew Turk

Table 1 Existing feature-based visual tracking systems. This compilation is not exhaustive

Reference	Objective/prior knowledge or additional inputs ^a	Detector ^b	Descriptor ^b	Matching, outlier removal, pose estimation ^c
Bleser and Stricker (2008)	tracking/3D model, IMU	FAST	patch, warped	SSD, Kalman filter
Carrera et al. (2007)	tracking/known target	Harris	SURF	UKF
Chekhlov et al. (2007)	SLAM/–	Shi-Tomasi	SIFT-like	UKF
Cheng et al. (2006)	odom./stereo, wheel odom., IMU	Harris or sim.	patch	NCC, RANSAC, LSE
Davison et al. (2007)	SLAM/initialization target	Shi-Tomasi ^d	patch (11 × 11), warped	NCC, EKF
DiVerdi et al. (2008)	panorama creation/–	Shi-Tomasi	Optical flow & SURF ^e	RANSAC (Horn 1987)
Eade and Drummond (2006b)	SLAM/–	FAST ^d	patch, warped	NCC, particle filter
Klein and Murray (2007)	SLAM/–	FAST	patch (8 × 8), warped	SSD, Tukey M-estimator
Lee and Höllerer (2008)	tracking/init. with user's hand	DoG	Optical flow & SIFT ^e	RANSAC, Kalman filter
Lepetit and Fua (2006)	tracking-by-det./known target	cf. reference	Randomized Trees	RANSAC, LSE, P- <i>n</i> -P
Nistér et al. (2004)	odometry/–	Harris	patch (11 × 11)	NCC, RANSAC
Özuysal et al. (2007)	tracking-by-det./known target	cf. reference	Ferns	RANSAC
Park et al. (2008)	tracking-by-det./known targets	not specified	Ferns	RANSAC, P- <i>n</i> -P
Se et al. (2002)	SLAM/trinocular camera	DoG	[scale, orientation]	LSE, Kalman filter
Skrypnik and Lowe (2004)	tracking/known scene	DoG	SIFT	RANSAC, non-lin. LSE
Taylor et al. (2009)	tracking-by-det./known target	FAST	trained histograms	PROSAC
Wagner et al. (2009)	tracking/known targets	FAST ^d	patch & reduced SIFT ^e	NCC, PROSAC, M-estim.
Wagner et al. (2010)	panorama creation/–	FAST ^d	patch (8 × 8), warped	NCC, M-estimator
Williams et al. (2007)	recovery for SLAM/–	FAST	Randomized lists	JCBB, EKF

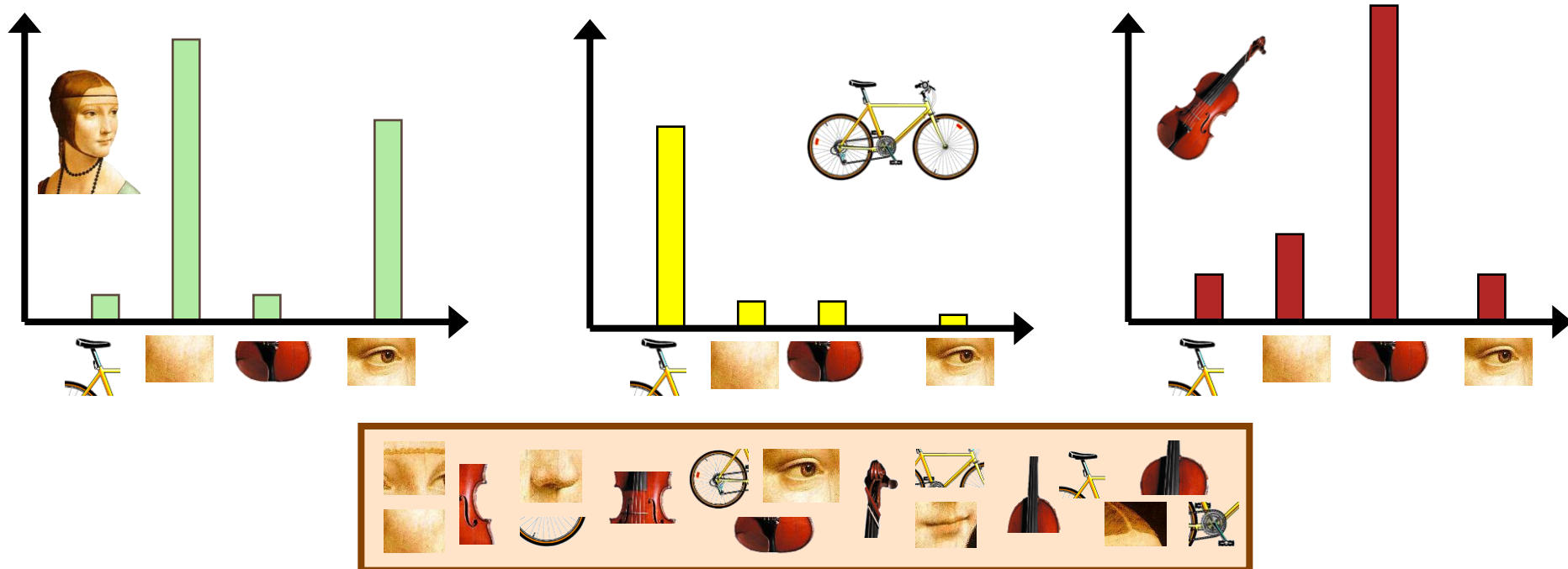
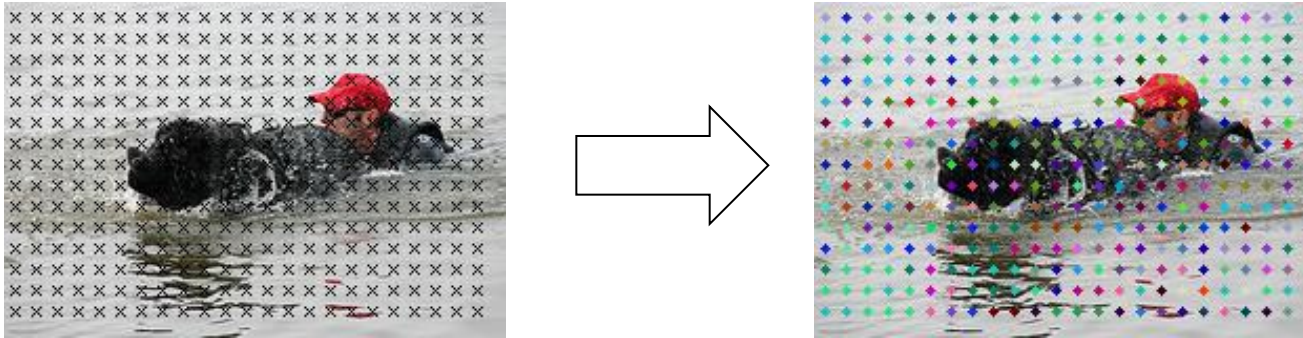
Visual Feature Clustering – the “Dictionary”

- Iterate between “assignment” and “center-updating” steps
 - Each sample is assigned to the most similar center
 - Update centers by averaging all assigned samples

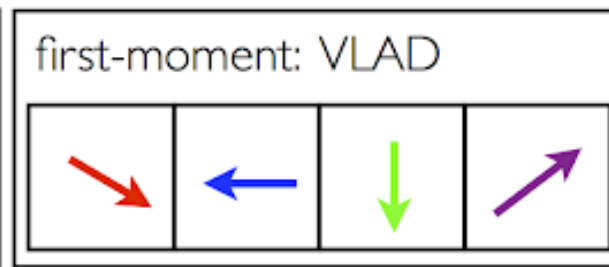
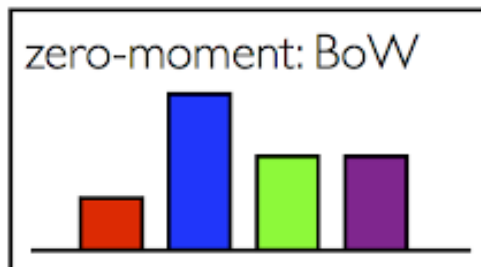
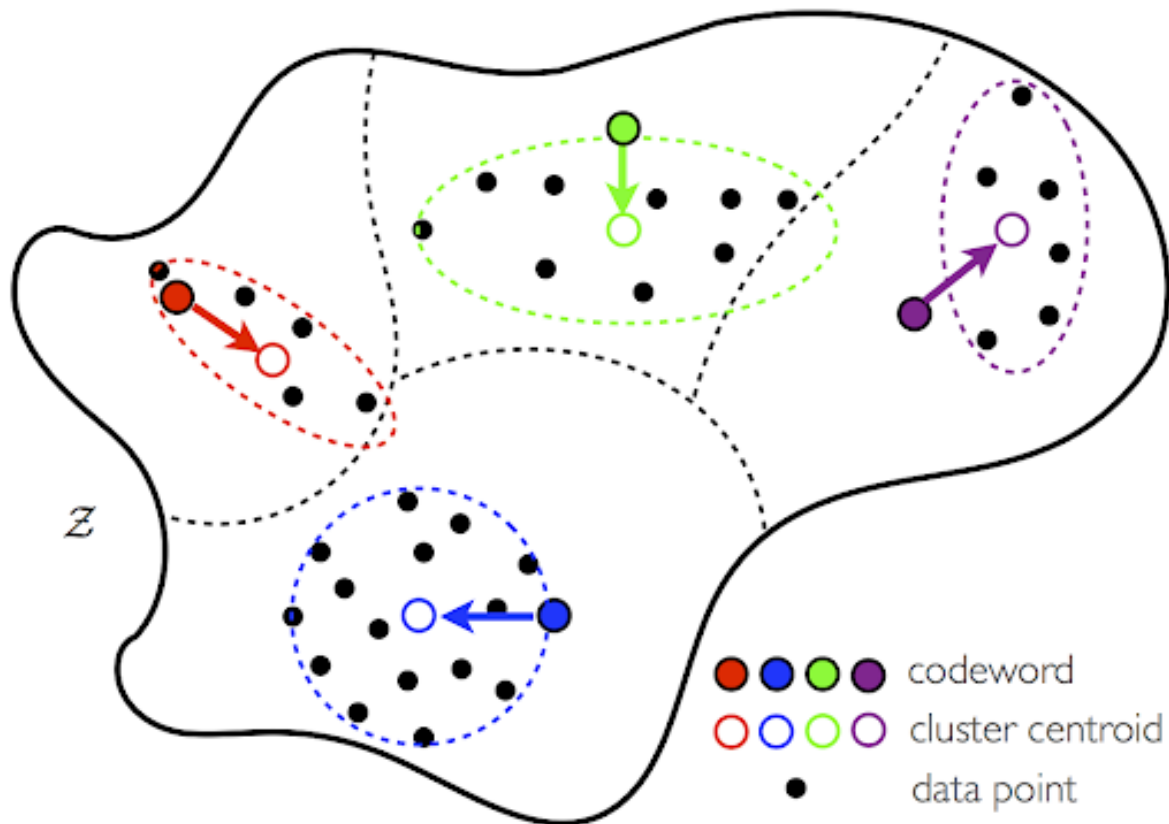


Local Feature Quantization

- Each SIFT descriptor is quantized into a visual word using the nearest cluster center.

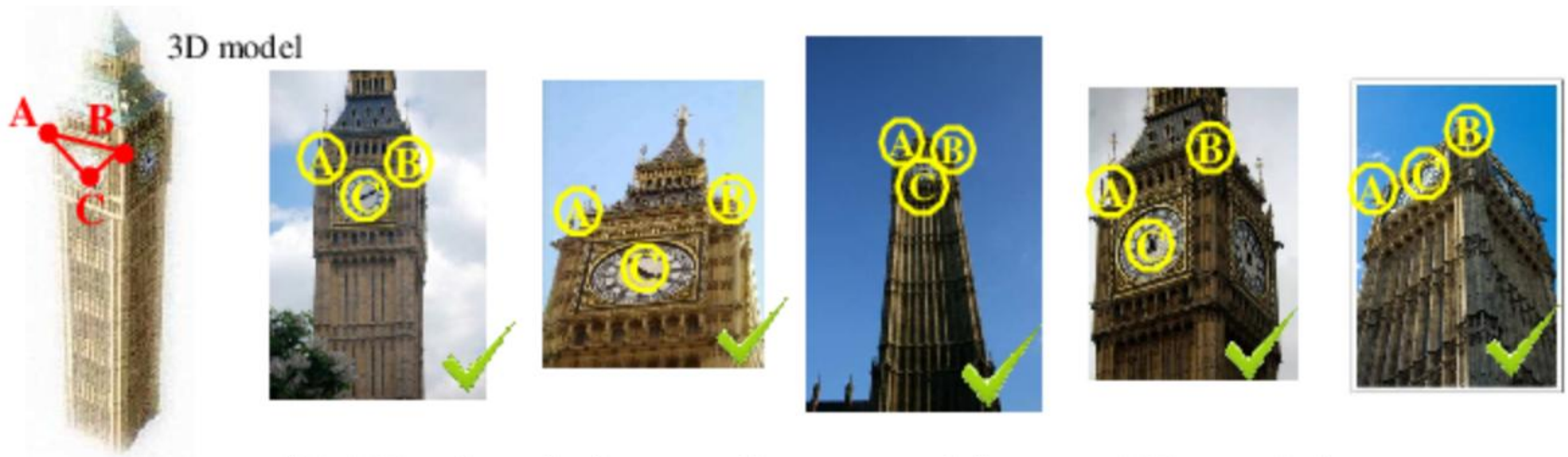


VLAD



Visual Phrases

- Bundling multiple local features for more discriminative visual pattern
- Often learned from data



Fisher Vector

- Extension from 1-D Moment to 2-D
- Further readings
 - Large-scale visual recognition: Novel patch aggregation mechanisms, CVPR 2012 Tutorial
 - FV derivation, <https://www.vlfeat.org/api/fisher-derivation.html>

How can we improve the BoW model

- Utilize spatial information
- Reduce quantization error
- The way to learn the dictionary
- The granularity of BoW
- More aggregation beyond voting
- ...

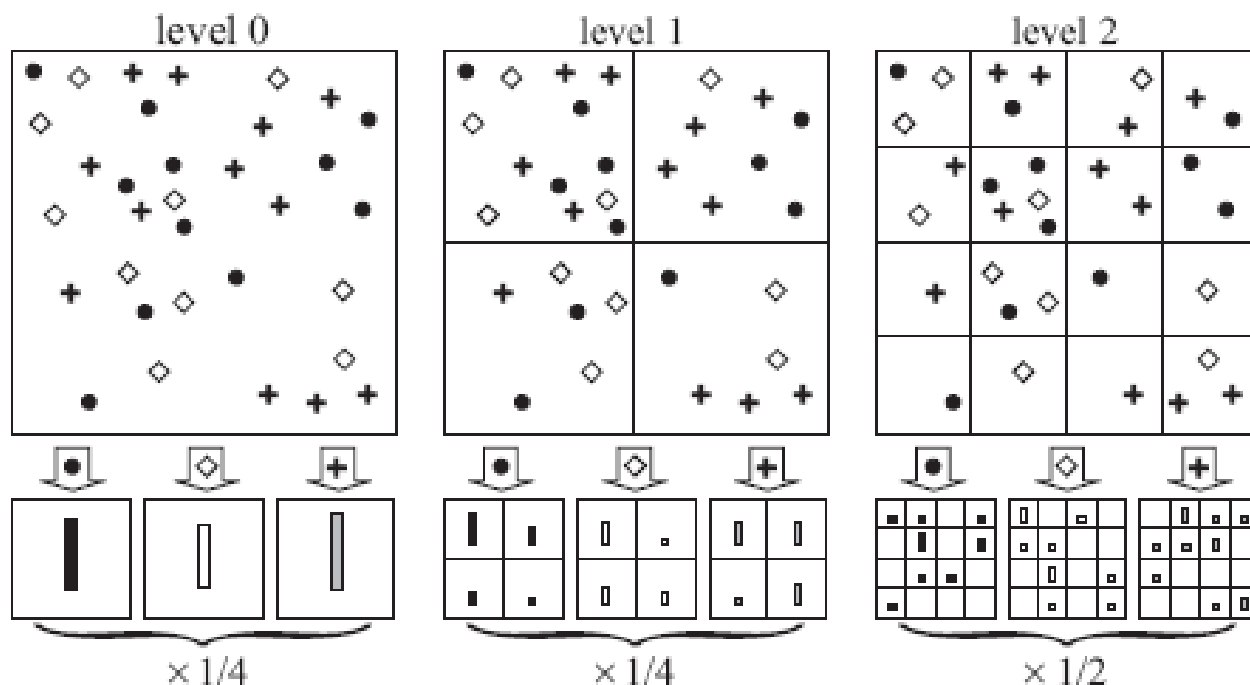
Today we will talk about

- **(Locality): Spatial Pyramid Matching & Pyramid Match Kernel**
- **(Aggregation): VLAD, Fisher Vector, Soft Assignment**
- **(Dictionary): Vocabulary Tree, Sparse Coding**

Spatial Pyramid Matching (SPM)

Spatial Pyramid Matching

- Adding Spatial Information to the BoW model
- Locally orderless representation at several levels of resolution



Svetlana Lazebnik et al., Beyond Bags of Features: Spatial Pyramid Matching for Recognizing Natural Scene Categories, CVPR 2006

Spatial Pyramid Matching

- Adding Spatial Information
- Locally orderless representation at several levels of resolution



Svetlana Lazebnik et al., Beyond Bags of Features: Spatial Pyramid Matching for Recognizing Natural Scene Categories, CVPR 2006

Caltech101 dataset

Fei-Fei et al. (2004)

http://www.vision.caltech.edu/Image_Datasets/Caltech101/Caltech101.html

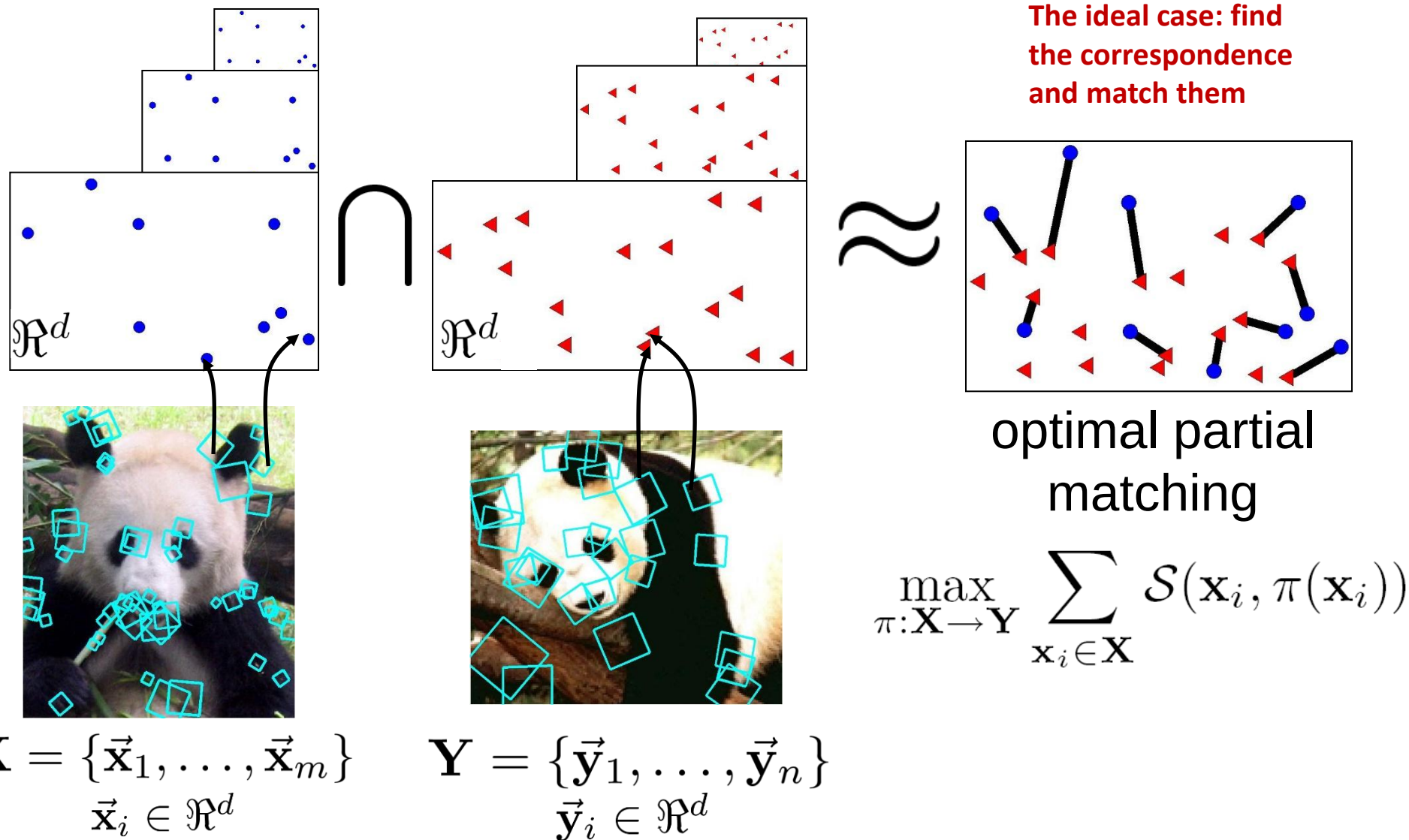


Multi-class classification results (30 training images per class)

	Weak features (16)		Strong features (200)	
Level	Single-level	Pyramid	Single-level	Pyramid
0	15.5 $\pm$ 0.9		41.2 $\pm$ 1.2	
1	31.4 $\pm$ 1.2	32.8 $\pm$ 1.3	55.9 $\pm$ 0.9	57.0 $\pm$ 0.8
2	47.2 $\pm$ 1.1	49.3 $\pm$ 1.4	63.6 $\pm$ 0.9	64.6 $\pm$ 0.8
3	52.2 $\pm$ 0.8	54.0 $\pm$ 1.1	60.3 $\pm$ 0.9	64.6 $\pm$ 0.7

Pyramid Match Kernel (PMK)

Motivation of Pyramid Match Kernel



Pyramid Match Kernel

- Pyramid match kernel aims to measure the similarity of a partial matching between two sets:
- Solution:
 - Place multi-dimensional, multi-resolution grid over point sets, and obtain multi-scale histograms

$$\Psi(\mathbf{X}) = [H_0(\mathbf{X}), \dots, H_L(\mathbf{X})]$$

- Consider points matched at finest resolution where they fall into same grid cell
- Approximate similarity between matched points with worst case similarity at given level

No explicit search for matches!

Pyramid Match Kernel

The pyramid match kernel measures similarity between point sets based on implicit correspondences found within this multi-resolution histogram space.

Approximate
partial match
similarity

$$K_{\Delta} = \sum_{i=0}^L w_i N_i$$

Number of newly matched pairs
at level i

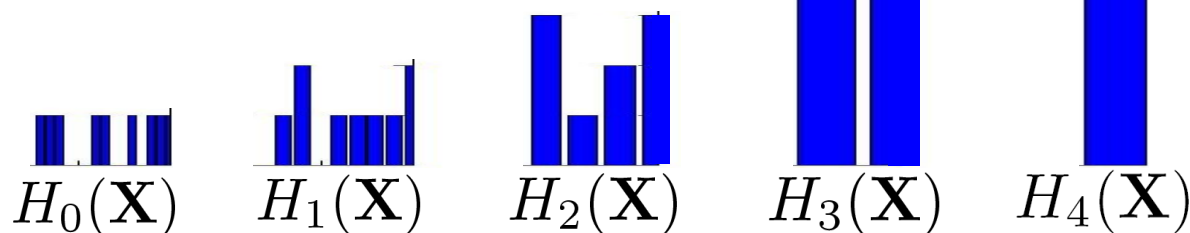
Measure of difficulty of a
match at level i

Building Multi-Scale Histograms

$$\mathbf{X} = \{\vec{\mathbf{x}}_1, \dots, \vec{\mathbf{x}}_m\}, \quad \vec{\mathbf{x}}_i \in \mathbb{R}^d$$



Histogram pyramid: level i
has bins of size 2^i

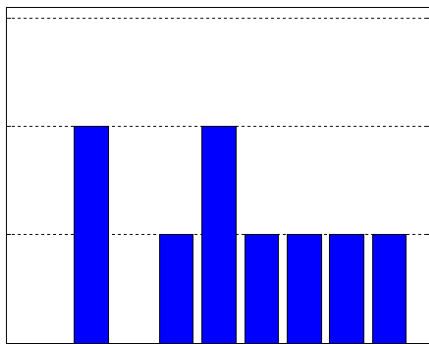
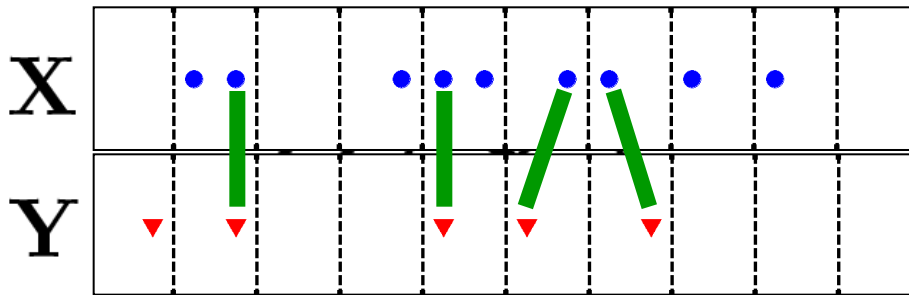


$$\Psi(\mathbf{X}) = [H_0(\mathbf{X}), \dots, H_L(\mathbf{X})]$$

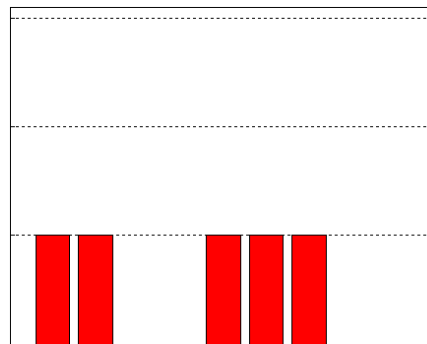
Counting Matches

Histogram intersection

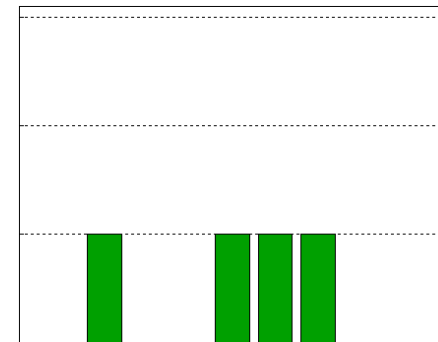
$$\mathcal{I}(H(\mathbf{X}), H(\mathbf{Y})) = \sum_{j=1}^r \min(H(\mathbf{X})_j, H(\mathbf{Y})_j)$$



$H(\mathbf{X})$

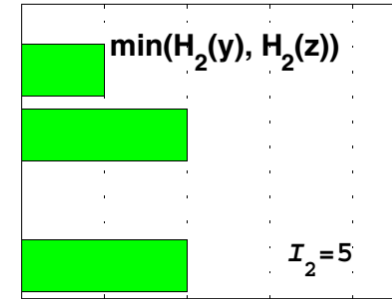
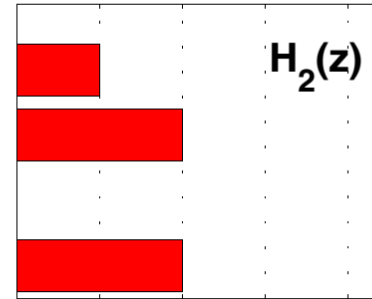
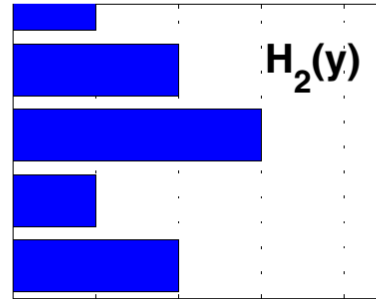
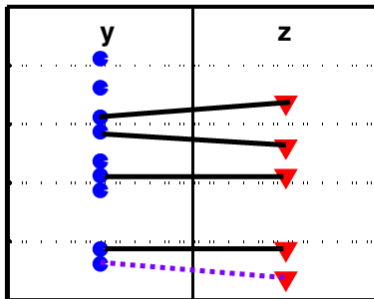
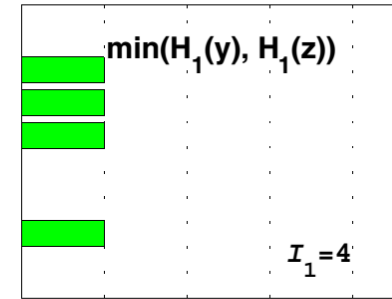
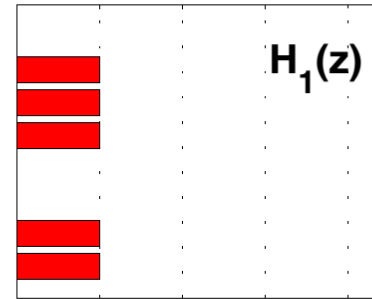
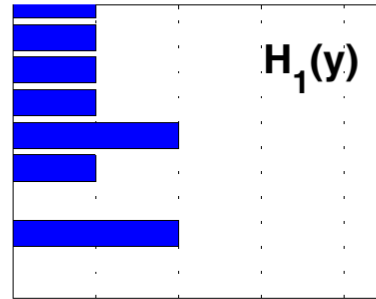
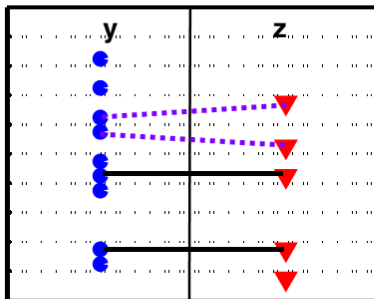
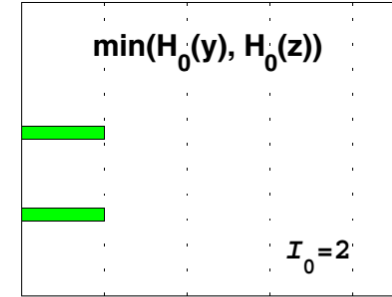
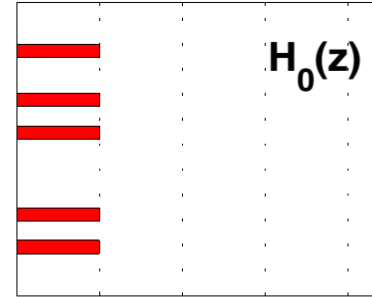
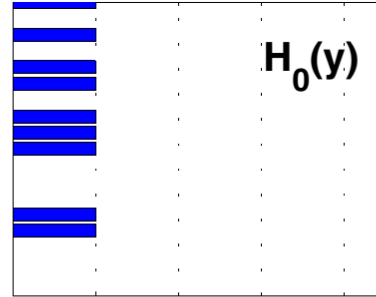
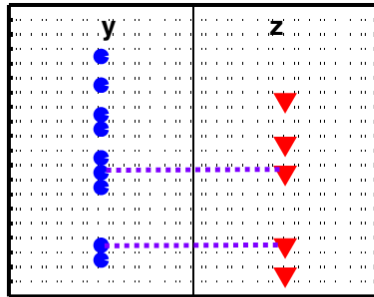


$H(\mathbf{Y})$



$$\mathcal{I}(H(\mathbf{X}), H(\mathbf{Y})) = 4$$

How to Merge Matches at All Levels?



(a) Point sets

(b) Histogram pyramids

(c) Intersections

Counting New Matches

Histogram intersection

$$\mathcal{I}(H(\mathbf{X}), H(\mathbf{Y})) = \sum_{j=1}^r \min(H(\mathbf{X})_j, H(\mathbf{Y})_j)$$

The key formula

$$N_i = \overbrace{\mathcal{I}(H_i(\mathbf{X}), H_i(\mathbf{Y}))}^{\text{matches at this level}} - \overbrace{\mathcal{I}(H_{i-1}(\mathbf{X}), H_{i-1}(\mathbf{Y}))}^{\text{matches at previous level}}$$

Difference in histogram intersections across levels counts *number of new pairs* matched

Pyramid Match Kernel

$$K_{\Delta} \left(\overbrace{\Psi(\mathbf{X}), \Psi(\mathbf{Y})}^{\text{histogram pyramids}} \right) =$$

$$\sum_{i=0}^L \frac{1}{2^i} \underbrace{\left(\mathcal{I}(H_i(\mathbf{X}), H_i(\mathbf{Y})) - \mathcal{I}(H_{i-1}(\mathbf{X}), H_{i-1}(\mathbf{Y})) \right)}_{\text{number of newly matched pairs at level } i}$$

↑
measure of difficulty of
a match at level i

- Weights inversely proportional to bin size
- Normalize kernel values to avoid favoring large sets

$$K_{\Delta}(\mathbf{P}, \mathbf{Q}) = \frac{1}{\sqrt{C}} \tilde{K}_{\Delta}(\mathbf{P}, \mathbf{Q}) \quad C = \tilde{K}_{\Delta}(\mathbf{P}, \mathbf{P}) \tilde{K}_{\Delta}(\mathbf{Q}, \mathbf{Q})$$

PMK Overview

Find maximum-weight matching (weight is inversely proportional to distance)

Original images

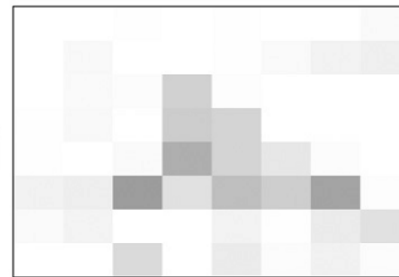


Feature histograms:

Level 3



$\cap$



$= \mathcal{I}_3$

Level 2



$\cap$

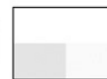


$= \mathcal{I}_2$

Level 1



$\cap$



$= \mathcal{I}_1$

Level 0



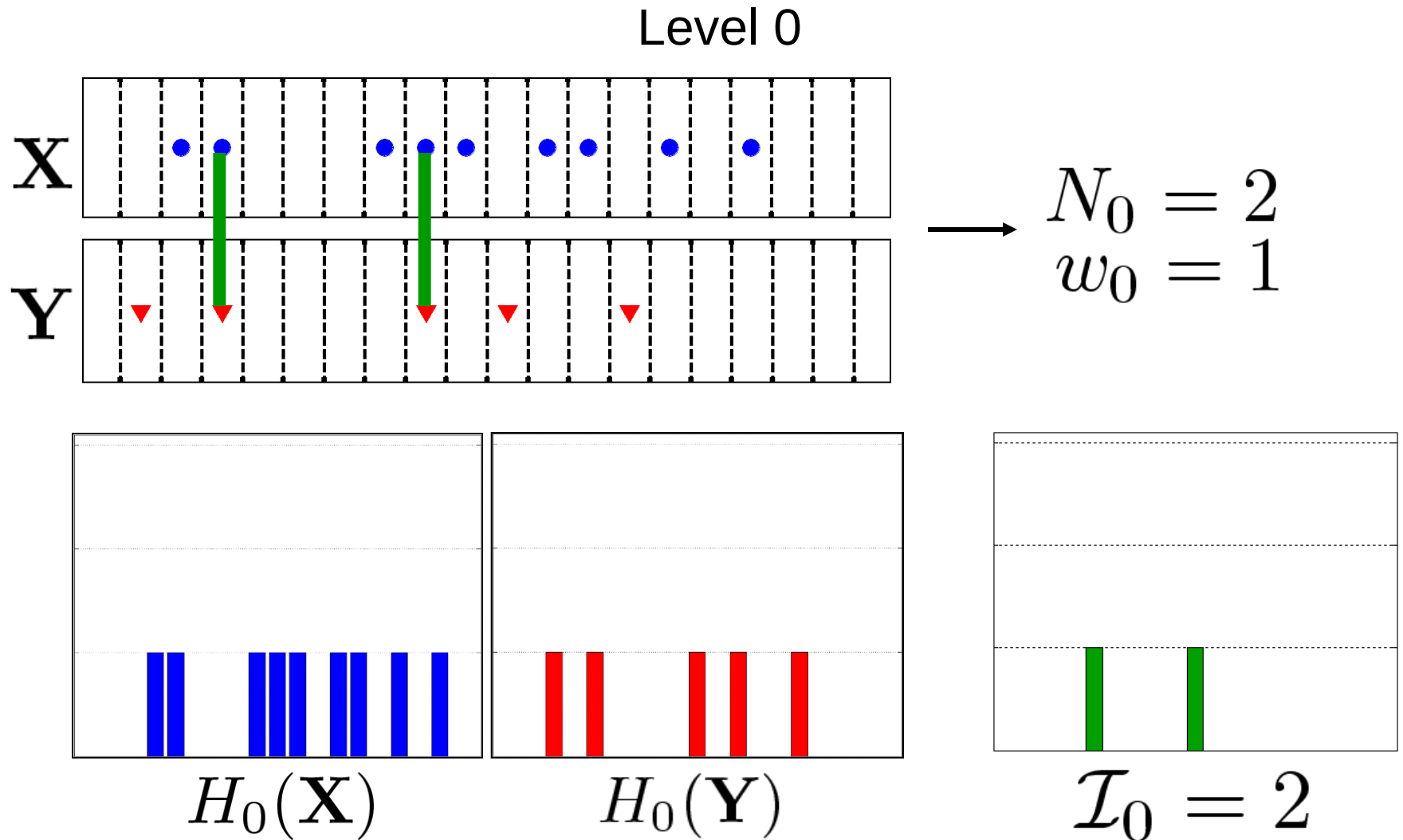
$\cap$



$= \mathcal{I}_0$

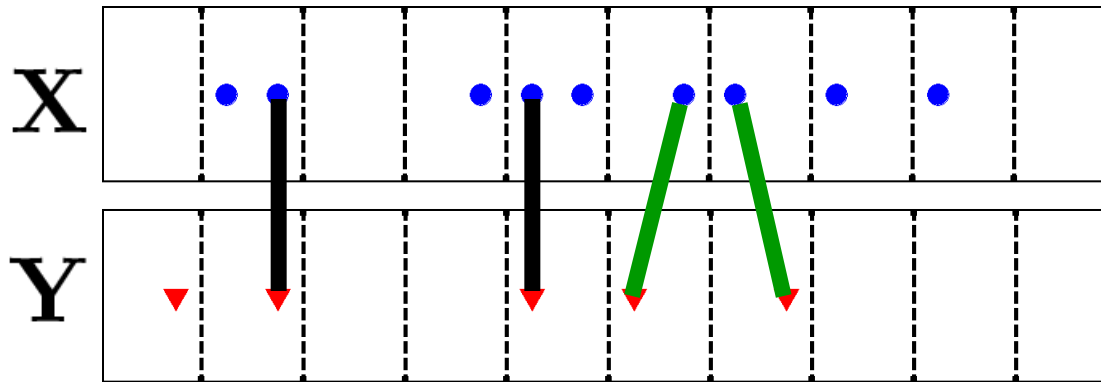
Total weight (value of *pyramid match kernel*): $\mathcal{I}_3 + \frac{1}{2}(\mathcal{I}_2 - \mathcal{I}_3) + \frac{1}{4}(\mathcal{I}_1 - \mathcal{I}_2) + \frac{1}{8}(\mathcal{I}_0 - \mathcal{I}_1)$

Example pyramid match

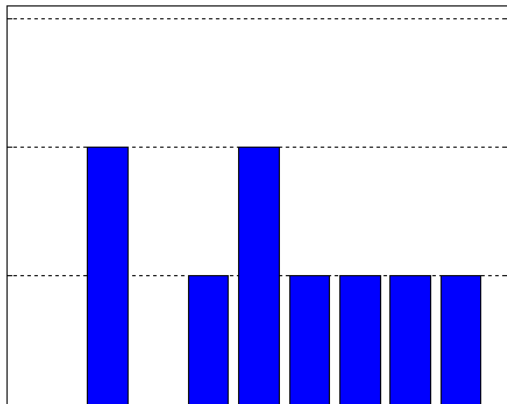


Example pyramid match

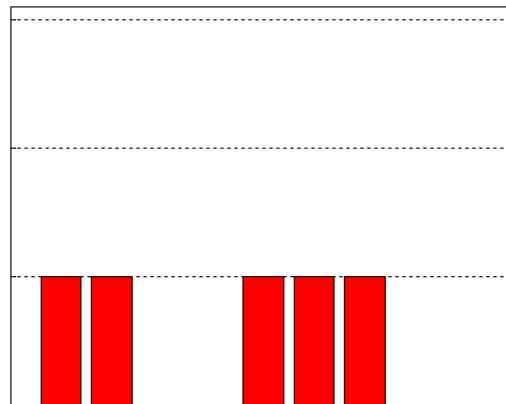
Level 1



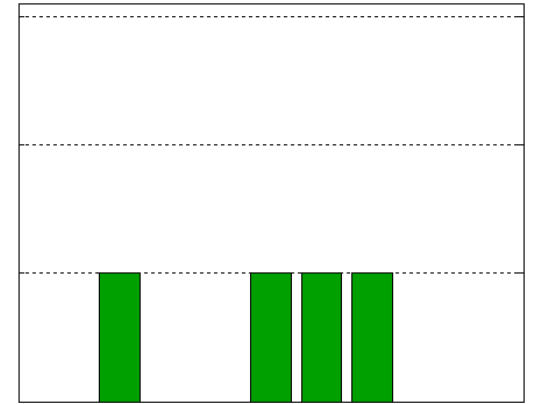
$$\rightarrow \begin{aligned} N_1 &= 4 - 2 = 2 \\ w_1 &= \frac{1}{2} \end{aligned}$$



$H_1(\mathbf{X})$



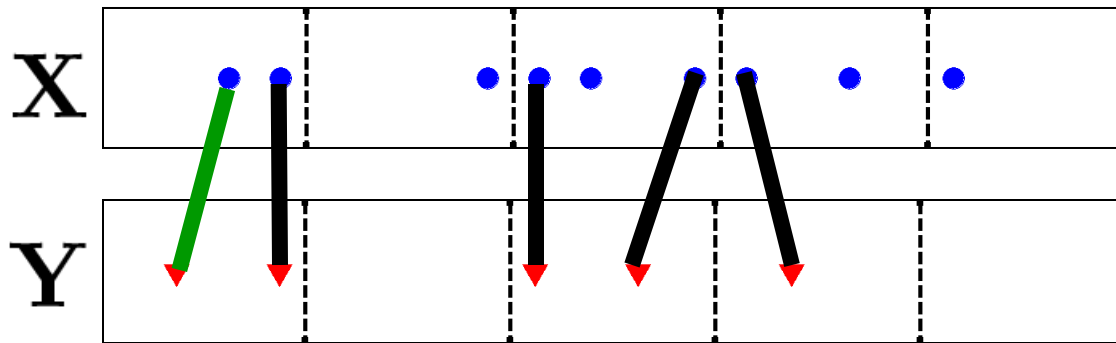
$H_1(\mathbf{Y})$



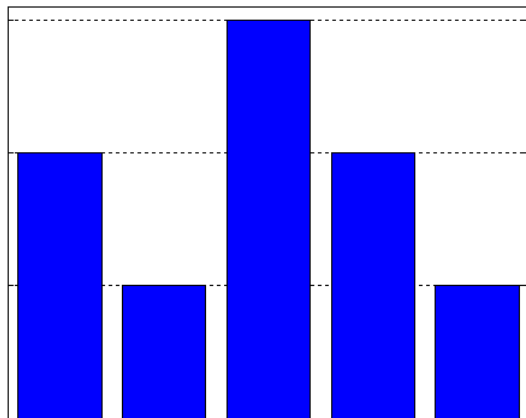
$\mathcal{I}_1 = 4$

Example pyramid match

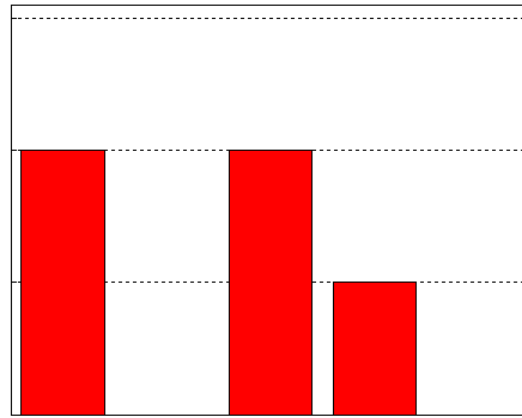
Level 2



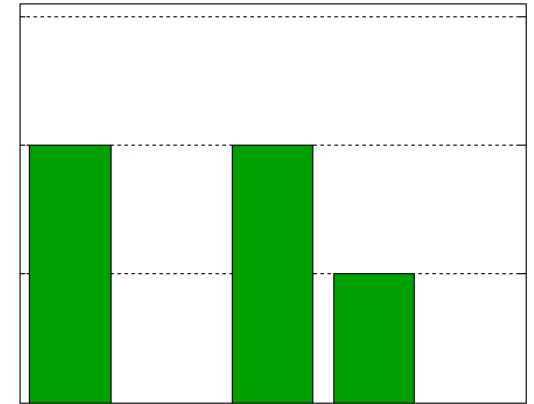
$$\begin{aligned} N_2 &= 5 - 4 = 1 \\ w_2 &= \frac{1}{4} \end{aligned}$$



$H_2(\mathbf{X})$



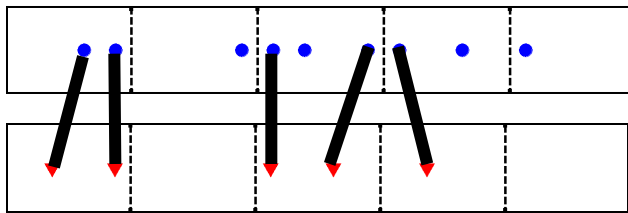
$H_2(\mathbf{Y})$



$\mathcal{I}_2 = 5$

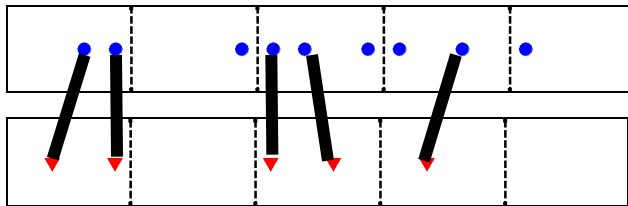
Example pyramid match

pyramid match



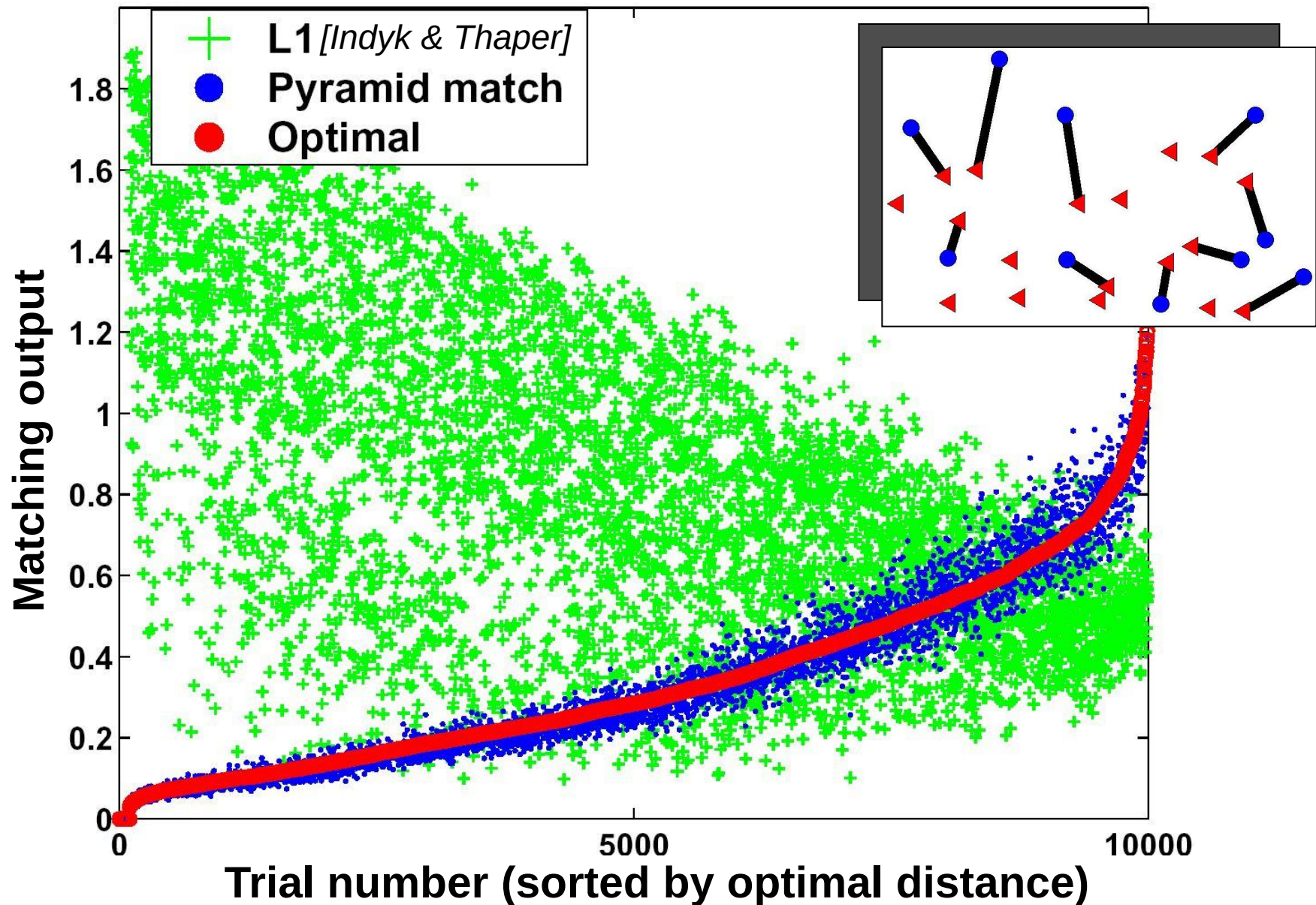
$$\begin{aligned}
 K_{\Delta} &= \sum_{i=0}^L w_i N_i \\
 &= 1(2) + \frac{1}{2}(2) + \frac{1}{4}(1) = 3.25
 \end{aligned}$$

optimal match



$$\begin{aligned}
 K &= \max_{\pi: \mathbf{X} \rightarrow \mathbf{Y}} \sum_{\mathbf{x}_i \in \mathbf{X}} \mathcal{S}(\mathbf{x}_i, \pi(\mathbf{x}_i)) \\
 &= 1(2) + \frac{1}{2}(3) = 3.5
 \end{aligned}$$

Approximation of the optimal partial matching



100 sets with 2D points, cardinalities vary between 5 and 100

Vocabulary Trees

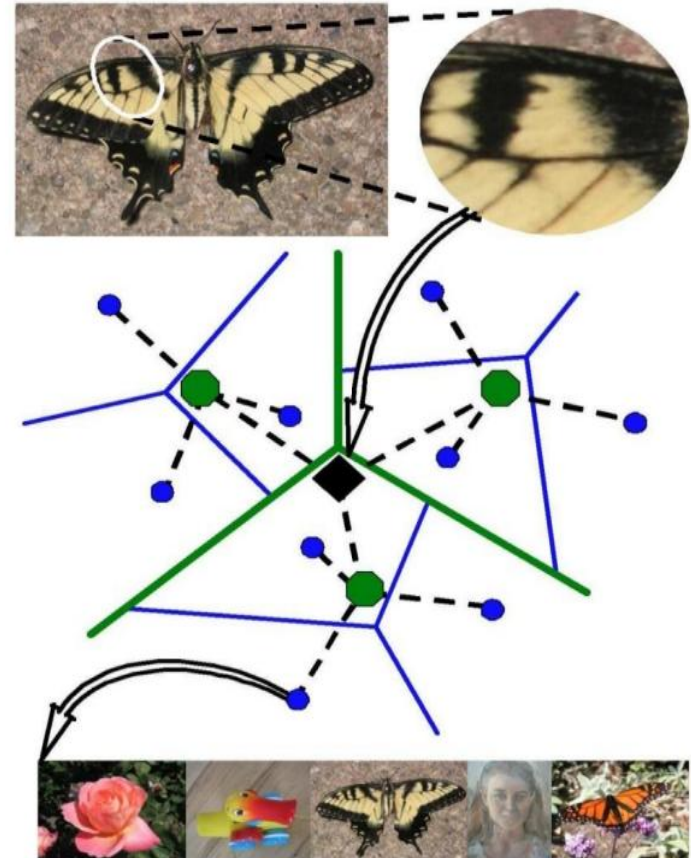
- **Vocabulary Tree** defined using an offline unsupervised (k-means) training stage.
- **Hierarchical scoring** based on term freq. inverse document freq. (TF-IDF).

- Number of the descriptor vectors of each image with a path along the node i (n_i query, m_i database)
- Number of images in the database with at least one descriptor vector path through the node i (N_i)
- Defining the relevance score

$$w_i = \ln \frac{N}{N_i} \quad \begin{matrix} q_i & = & n_i w_i \\ d_i & = & m_i w_i \end{matrix} \quad s(q, d) = \left\| \frac{q}{\|q\|} - \frac{d}{\|d\|} \right\|$$

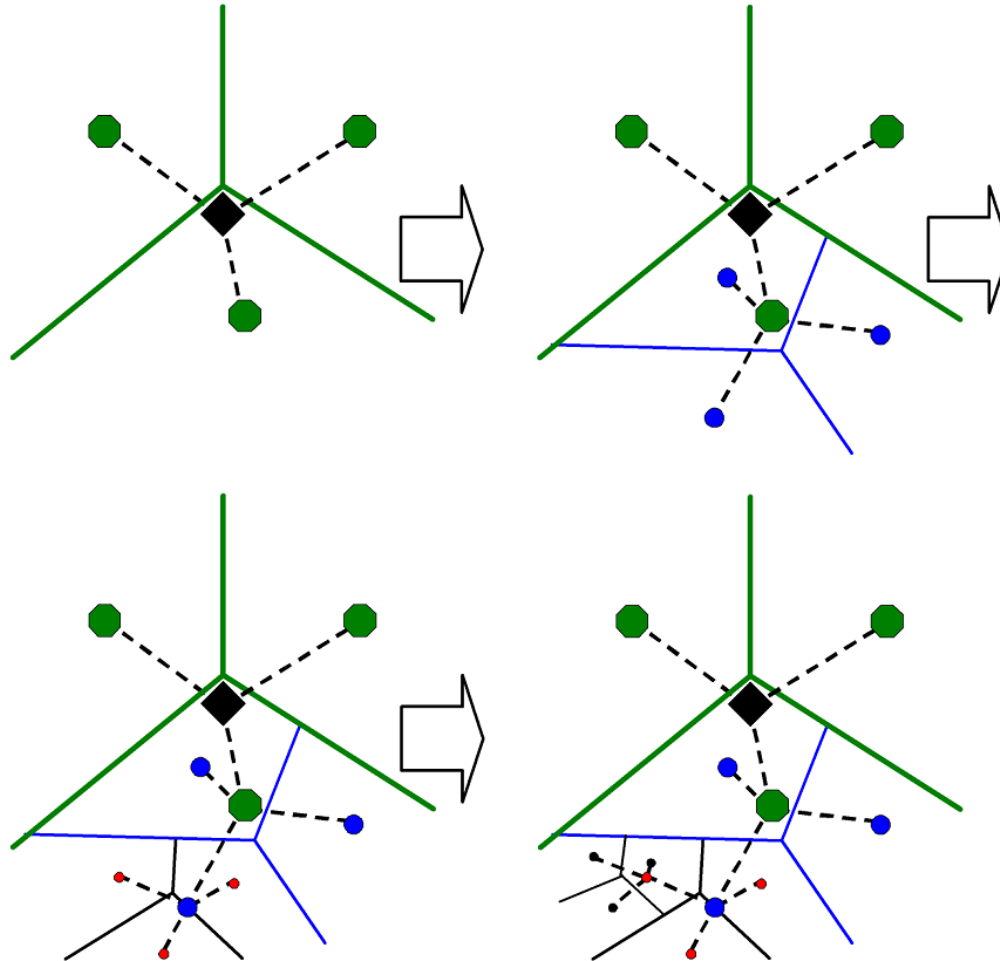
- **Implementation of Scoring**
 - Every node is associated with an inverted file
 - Decrease the fraction of images in the database that have to be explicitly considered for a query
- **Hierarchical k-means**
 - K-means tree of height h (levels)
- Determining the path of a descriptor means performing kh dot products.

Nister & Stewenius, 2006



Vocabulary Tree

- Learn the dictionary



Vocabulary Tree

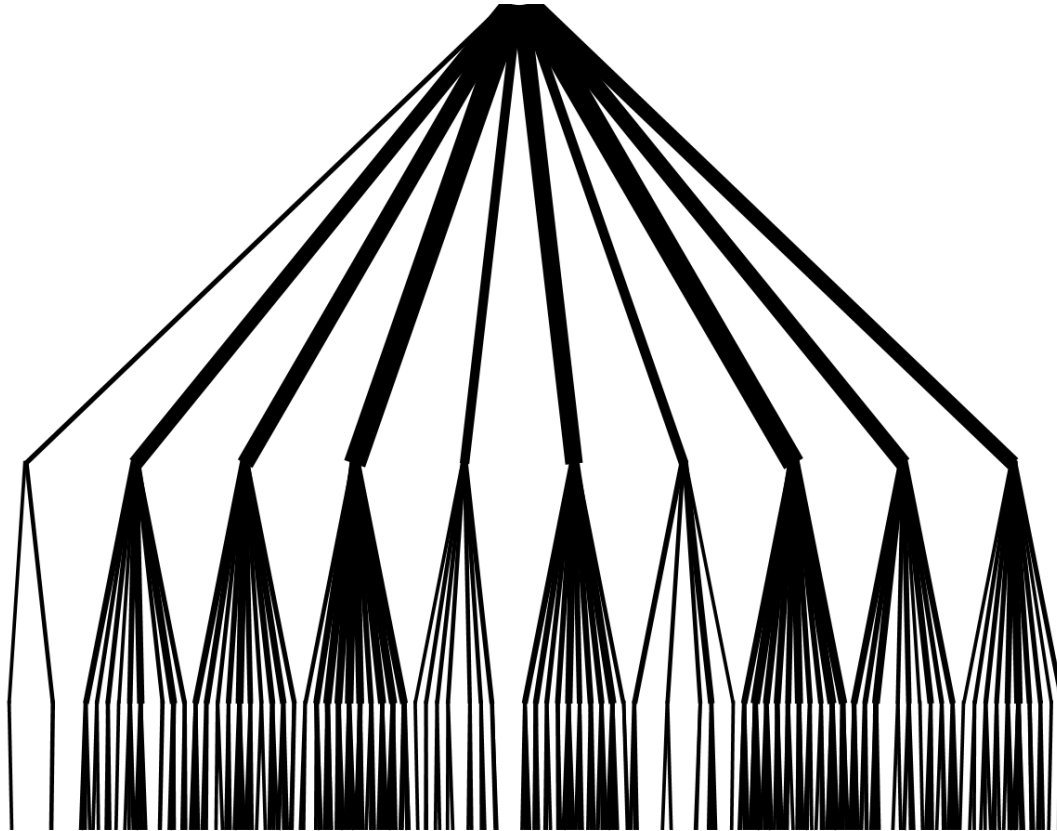
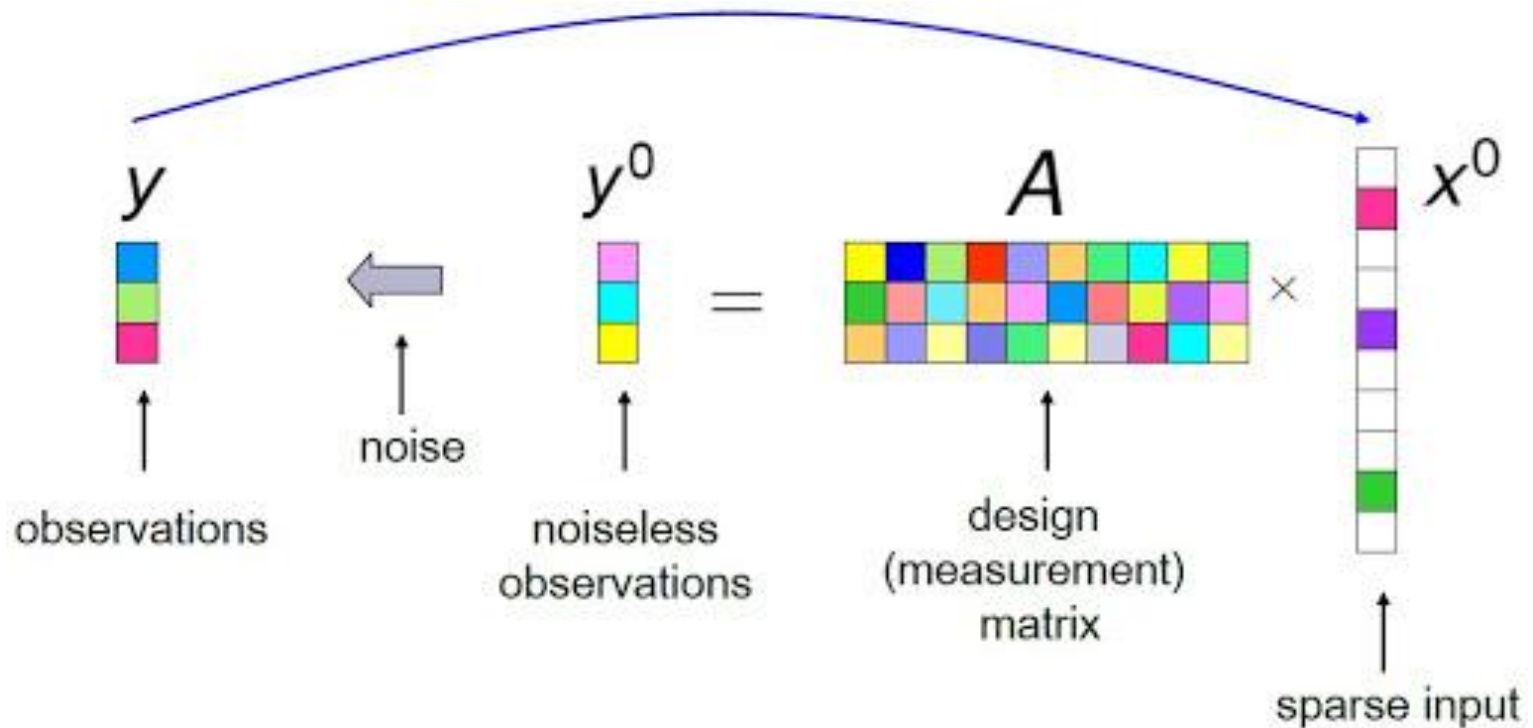


Figure 3. Three levels of a vocabulary tree with branch factor 10 populated to represent an image with 400 features.

Sparse Coding

- Jointly learning visual dictionary and image codes



Sparse Coding

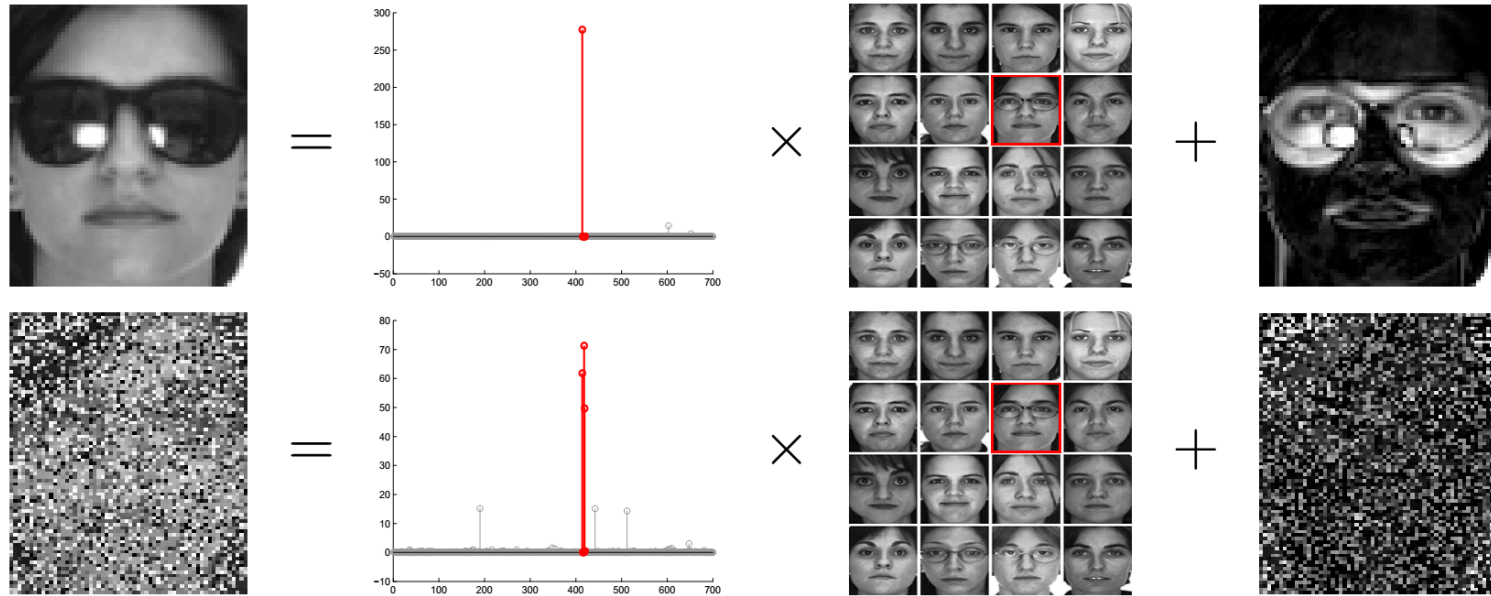
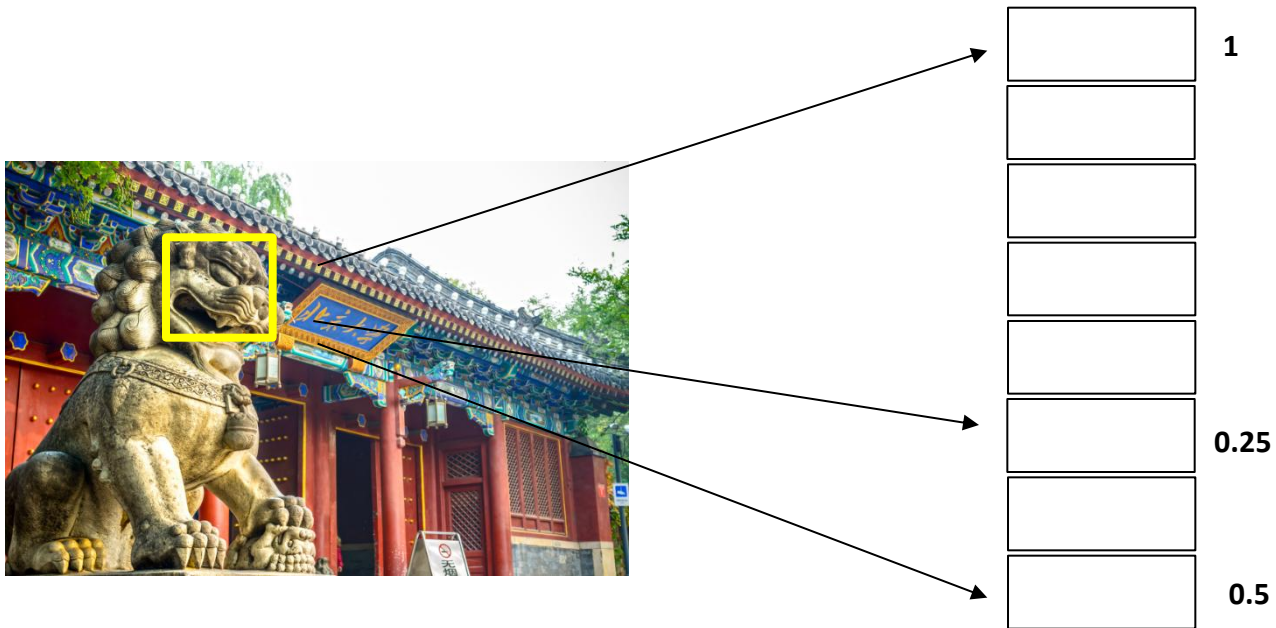


Fig. 1. **Overview of the face recognition approach.** The method represents a test image (left), which is potentially occluded (top) or corrupted (bottom), as a sparse linear combination of all the training images (middle) plus sparse errors (right) due to occlusion or corruption [71]. Red (darker) coefficients correspond to training images of the correct individual. The algorithm determines the true identity (indicated with a red box at second row and third column) from 700 training images of 100 individuals (7 each) in the standard AR face database.

Soft Assignment

- Size of visual dictionary v.s. Quantization error
- Each local feature votes for multiple bins/words



Yu-Gang Jiang, Chong-Wah Ngo, Jun Yang: Towards optimal bag-of-features for object categorization and semantic video retrieval. CIVR 2007

Soft Assignment

- The improvement is striking!

Table 3: The mean InfAP of semantic video retrieval on TRECVID-2006 dataset using different weighting schemes and vocabulary sizes. The best result at each vocabulary size is shown in bold.

Vocabulary size	Weighting schemes			
	<i>binary</i>	<i>tf</i>	<i>tf-idf</i>	<i>soft-weighting</i>
500	0.048	0.088	0.081	0.110
1,000	0.076	0.082	0.078	0.105
5,000	0.082	0.083	0.089	0.100
10,000	0.083	0.090	0.096	0.111

Table 2: The mean EER of object classification on PASCAL-2005 dataset using different weighting schemes and vocabulary sizes. The best result at each vocabulary size is shown in bold.

Vocabulary size	Weighting schemes			
	<i>binary</i>	<i>tf</i>	<i>tf-idf</i>	<i>soft-weighting</i>
500	0.906	0.924	0.928	0.922
1,000	0.913	0.925	0.929	0.931
2,000	0.916	0.913	0.914	0.935
5,000	0.921	0.917	0.913	0.931
10,000	0.904	0.902	0.908	0.927

Related Tutorials

- CVPR-2010 [Tutorial on Sparse Coding and Dictionary Learning for Image Analysis](#), by Francis Bach (INRIA), Julien Mairal(INRIA), Jean Ponce (Ecole Normale Supérieure), and Guillermo Sapiro (University of Minnesota).
- ICCV-2009 [Tutorial on Recognizing and Learning Object Categories](#), by Li Fei-Fei (Stanford), Rob Fergus (NYU), and Antonio Torralba (MIT)
- ECCV-2010 [Tutorial on Feature Learning for Image Classification](#), by Kai Yu (NEC), and Andrew Ng (Stanford)

Reference

- David Nistér, Henrik Stewénus: Scalable Recognition with a Vocabulary Tree. CVPR (2) 2006: 2161 -2168
- Herve Jegou, Matthijs Douze, Cordelia Schmid: Improving Bag-of-Features for Large Scale Image Search. International Journal of Computer Vision 87(3): 316-336 (2010)
- Florent Perronnin, Jorge Sánchez, Thomas Mensink: Improving the Fisher Kernel for Large-Scale Image Classification. ECCV (4) 2010: 143-156